

# 第2章 电阻电路的等效变换

## 本章重点

|     |                            |
|-----|----------------------------|
| 2.1 | 引言                         |
| 2.2 | 电路的等效变换                    |
| 2.3 | 电阻的串联和并联                   |
| 2.4 | 电阻的Y形连接和 $\Delta$ 形连接的等效变换 |
| 2.5 | 电压源、电流源的串联和并联              |
| 2.6 | 实际电源的两种模型及其等效变换            |
| 2.7 | 输入电阻                       |

## ● 重点:

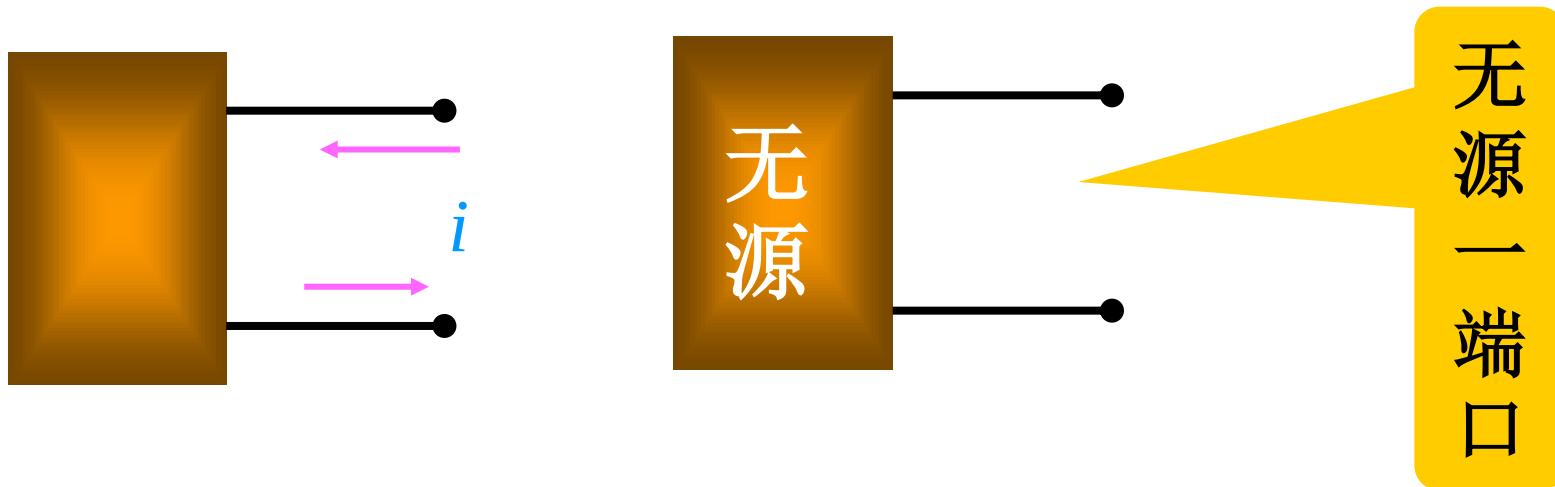
1. 电路等效的概念;
2. 电阻的串、并联;
3. 电阻的Y— $\Delta$  变换;
4. 电压源和电流源的等效变换;

电阻电路 → 仅由电源和线性电阻构成的电路

分析方法 → ①欧姆定律和基尔霍夫定律是  
分析电阻电路的依据；  
②等效变换的方法, 也称化简的  
方法。

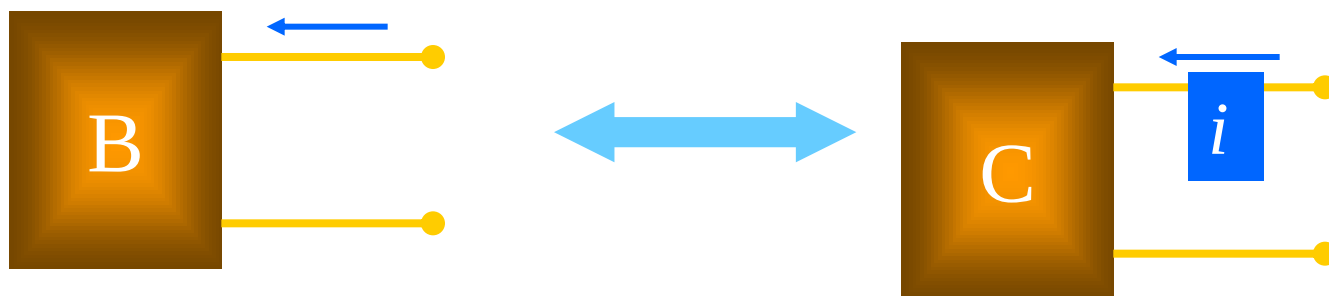
## 2.2 电路的等效变换

任何一个复杂的电路，向外引出两个端钮，且从一个端子流入的电流等于从另一端子流出的电流，则称这一电路为二端网络（或一端口网络）。

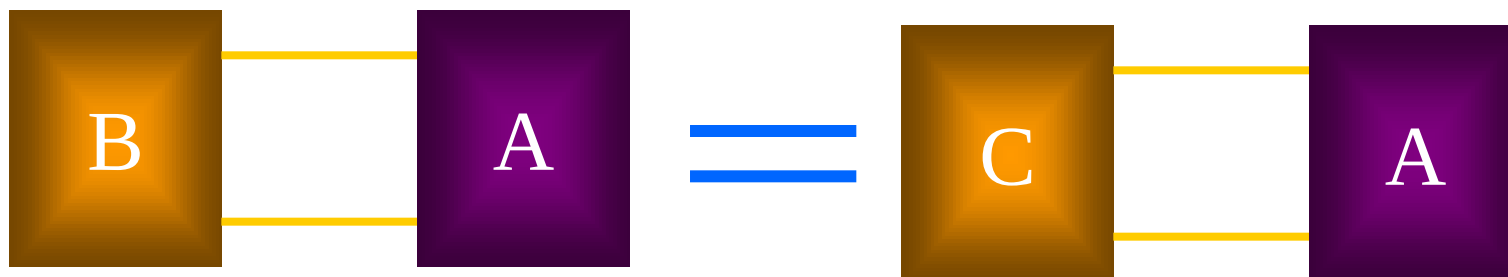


## 2. 两端电路等效的概念

两个两端电路，端口具有相同的电压、电流关系，则称它们是等效的电路。



对A电路中的电流、电压和功率而言，满足：





→ 两电路具有相同的VCR;

## ②电路等效变换的对象:

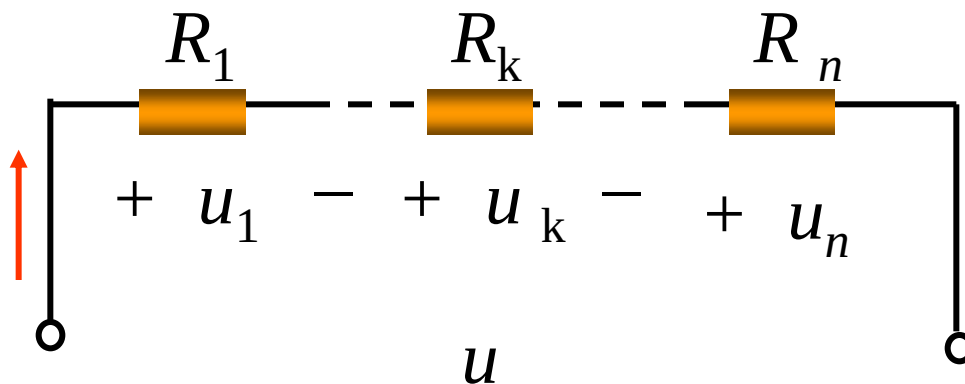
→ 未变化的外电路A中的电压、电流和功率;  
(即对外等效, 对内不等效)

## ③电路等效变换的目的:

→ 化简电路, 方便计算。

## 2.3 电阻的串联和并联

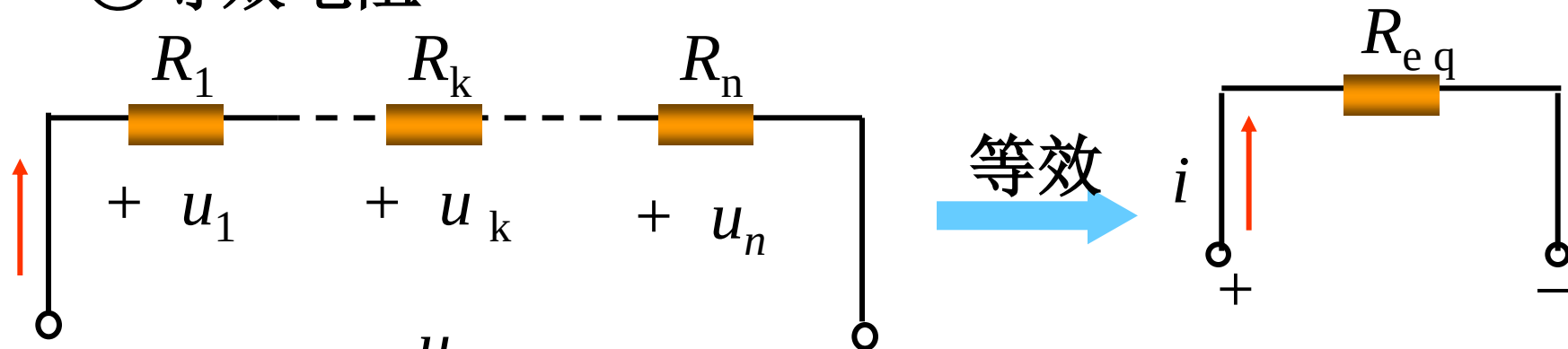
### ① 电路特点



- (a) 各电阻顺序连接，流过同一电流 (KCL);
- (b) 总电压等于各串联电阻的电压之和 (KVL)。

$$u = u_1 + \cdots + u_k + \cdots + u_n$$

## ②等效电阻



由欧姆定律 <sup>$u$</sup>

$$u = R_1 i + \cdots + R_k i + \cdots + R_n i = (R_1 + \cdots + R_n) i = R_{eq} i$$

$$R_{eq} = R_1 + \cdots + R_k + \cdots + R_n = \sum_{k=1}^n R_k > R_k$$

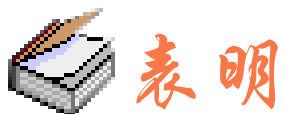


结论

串联电路的总电阻等于各分电阻之和。

### ③串联电阻的分压

$$u_k = R_k i = R_k \frac{u}{R_{eq}} = \frac{R_k}{R_{eq}} u < u$$



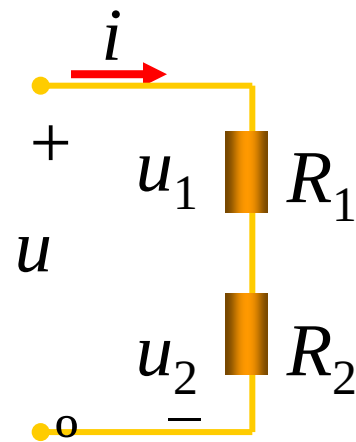
表明

电压与电阻成正比，因此串联电阻电路可作分压电路。

两个电阻的分压：

$$u_1 = \frac{R_1}{R_1 + R_2} u$$

$$u_2 = \frac{R_2}{R_1 + R_2} u$$



#### ④功率

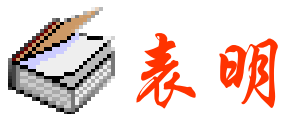
$$p_1=R_1i^2, \quad p_2=R_2i^2, \quad \dots, \quad p_n=R_ni^2$$

$$p_1:p_2:\dots:p_n=R_1:R_2:\dots:R_n$$

#### 总功率

$$p=R_{\text{eq}}i^2=(R_1+R_2+\dots+R_n)i^2$$

$$=R_1i^2+R_2i^2+\dots+R_ni^2$$



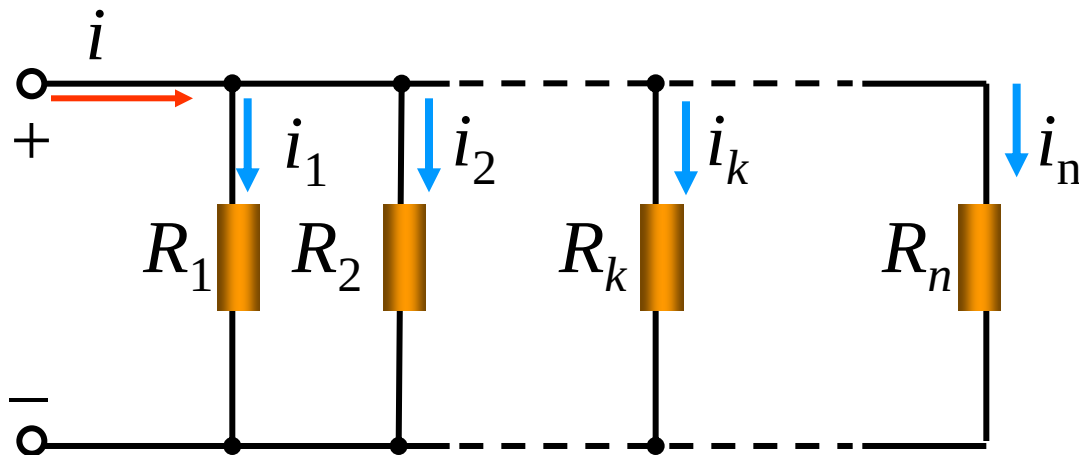
表明

$$=p_1+p_2+\dots+p_n$$

- ① 电阻串联时，各电阻消耗的功率与电阻大小成正比；
- ② 等效电阻消耗的功率等于各串联电阻消耗功率的总和。

## 2. 电阻并联

### ① 电路特点

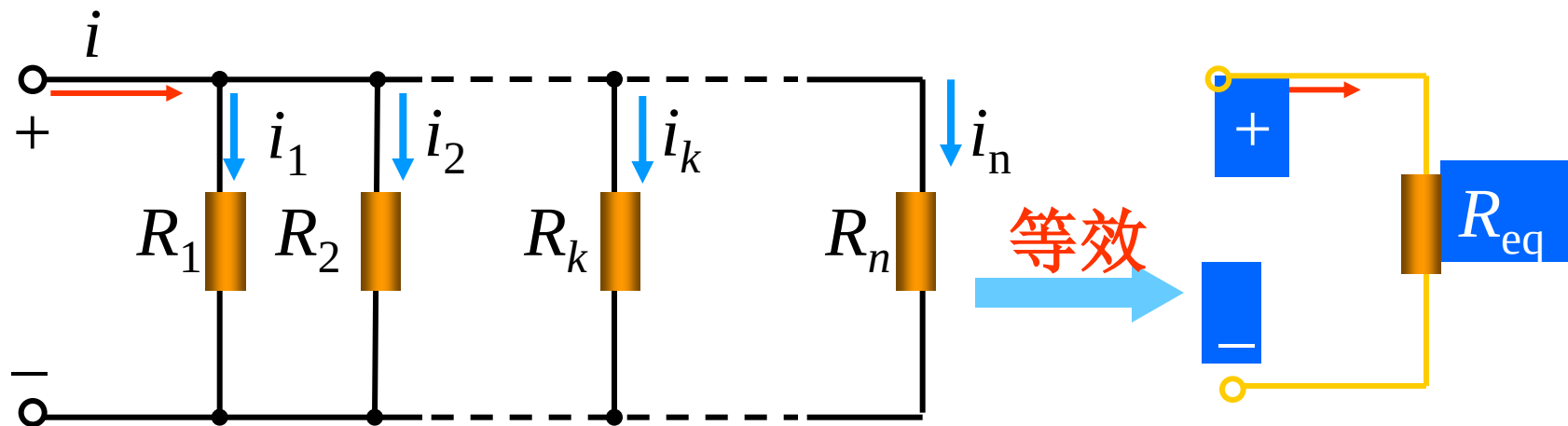


(a) 各电阻两端为同一电压 (KVL);

(b) 总电流等于流过各并联电阻的电流之和(KCL)。

$$i = i_1 + i_2 + \dots + i_k + \dots + i_n$$

## ②等效电阻



由KCL: 
$$i = i_1 + i_2 + \dots + i_k + \dots + i_n$$
$$= u/R_1 + u/R_2 + \dots + u/R_n$$
$$= u(1/R_1 + 1/R_2 + \dots + 1/R_n) = uG_{eq}$$
$$G_{eq} = G_1 + G_2 + \dots + G_n = \sum_{k=1}^n G_k > G_k$$



结论

等效电导等于并联的各电导之和。

$$\frac{1}{R_{eq}} = G_{eq} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n} \quad \text{即} \quad R_{eq} < R_k$$

### ③ 并联电阻的分流

电流分配与  
电导成正比

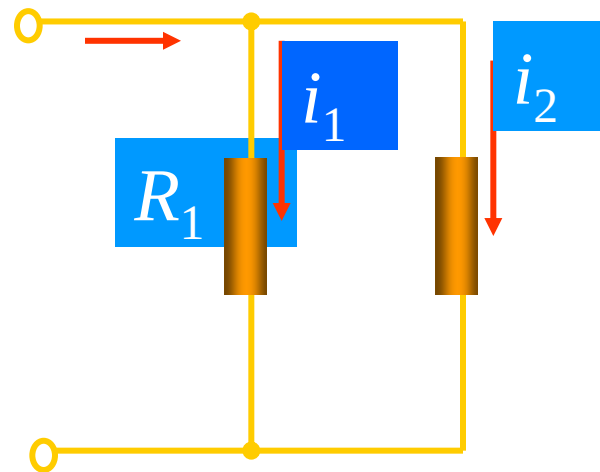
$$\frac{i_k}{i} = \frac{u / R_k}{u / R_{eq}} = \frac{G_k}{G_{eq}} \quad \rightarrow \quad i_k = \frac{G_k}{G_{eq}} i$$

## 两电阻的分流：

$$R_{eq} = \frac{1/R_1 \cdot 1/R_2}{1/R_1 + 1/R_2} = \frac{R_1 R_2}{R_1 + R_2}$$

$$i_1 = \frac{1/R_1}{1/R_1 + 1/R_2} i = \frac{R_2 i}{R_1 + R_2}$$

$$i_2 = \frac{1/R_2}{1/R_1 + 1/R_2} i = \frac{R_1 i}{R_1 + R_2} = (i - i_1)$$



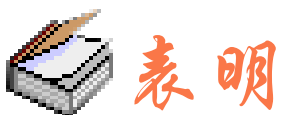
#### ④功率

$$p_1=G_1u^2, \quad p_2=G_2u^2, \quad \dots, \quad p_n=G_nu^2$$

$$p_1:p_2:\dots:p_n=G_1:G_2:\dots:G_n$$

总功率

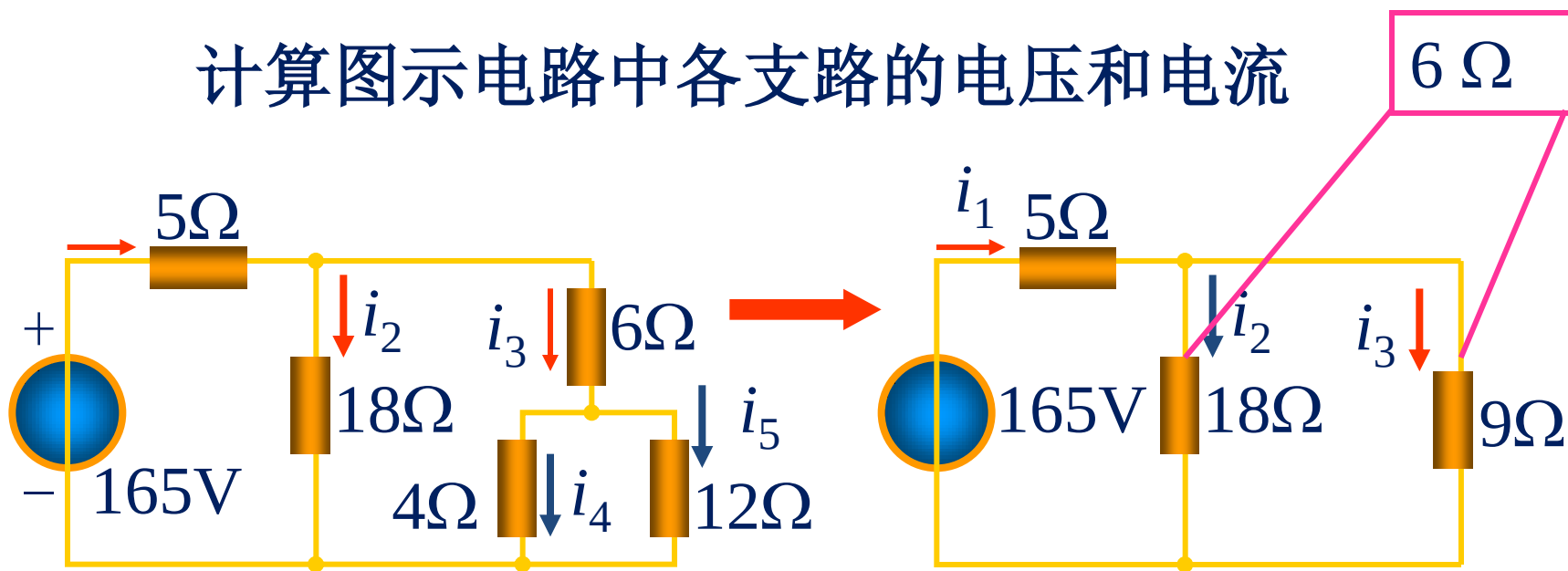
$$\begin{aligned} p &= G_{\text{eq}}u^2 = (G_1 + G_2 + \dots + G_n)u^2 \\ &= G_1u^2 + G_2u^2 + \dots + G_nu^2 \\ &= p_1 + p_2 + \dots + p_n \end{aligned}$$



- ①电阻并联时，各电阻消耗的功率与电阻大小成反比；
- ②等效电阻消耗的功率等于各并联电阻消耗功率的总和

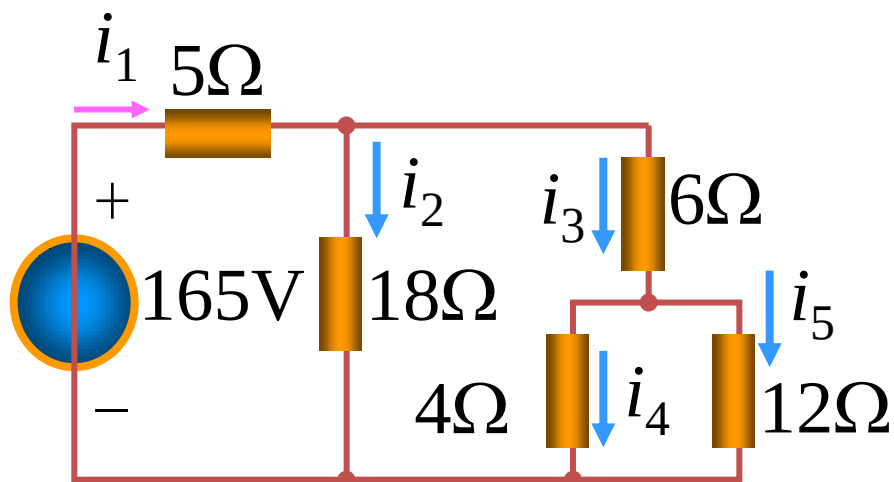
电路中有电阻的串联，又有电阻的并联，这种连接方式称电阻的串并联。

计算图示电路中各支路的电压和电流



$$i_1 = 165 / 11 = 15 \text{ A}$$

$$u_2 = 6i_1 = 6 \times 15 = 90 \text{ V}$$



$$i_2 = 90/18 = 5\text{A}$$

$$u_3 = 6i_3 = 6 \times 10 = 60\text{V}$$

$$i_3 = 15 - 5 = 10\text{A}$$

$$u_4 = 3i_3 = 30\text{V}$$

$$i_4 = 30/4 = 7.5\text{A}$$

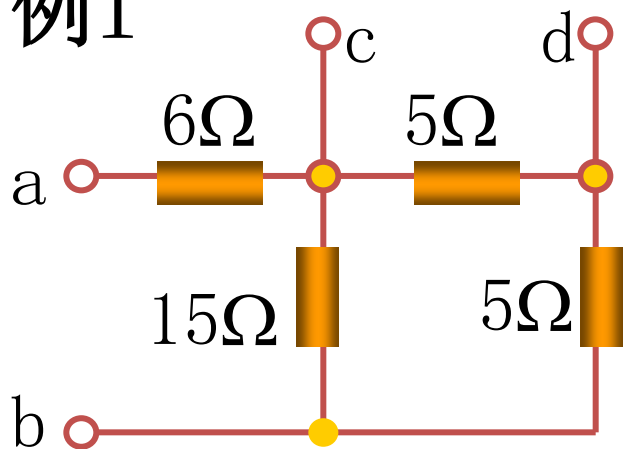
$$i_5 = 10 - 7.5 = 2.5\text{A}$$

从以上例题可得求解串、并联电路的一般步骤：

- ① 求出等效电阻或等效电导；
- ② 应用欧姆定律求出总电压或总电流；
- ③ 应用欧姆定律或分压、分流公式求各电阻上的电流和电压

以上的关键在于识别各电阻的串联、并联关系！

例1



求：  $R_{ab}$  ,  $R_{cd}$

$$R_{ab} = (5 + 5) // 15 + 6 = 12\Omega$$

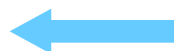
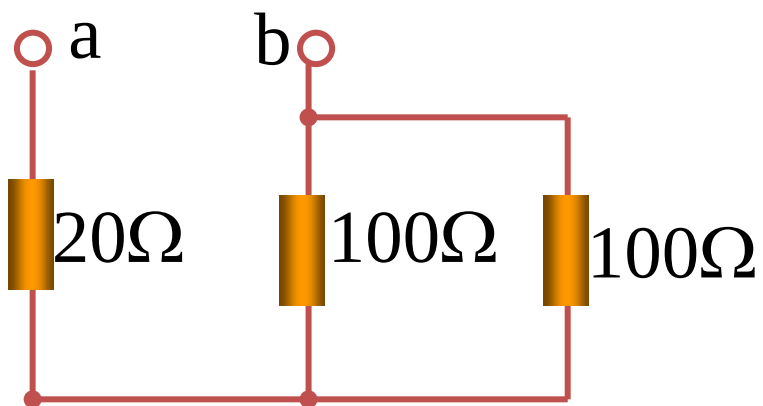
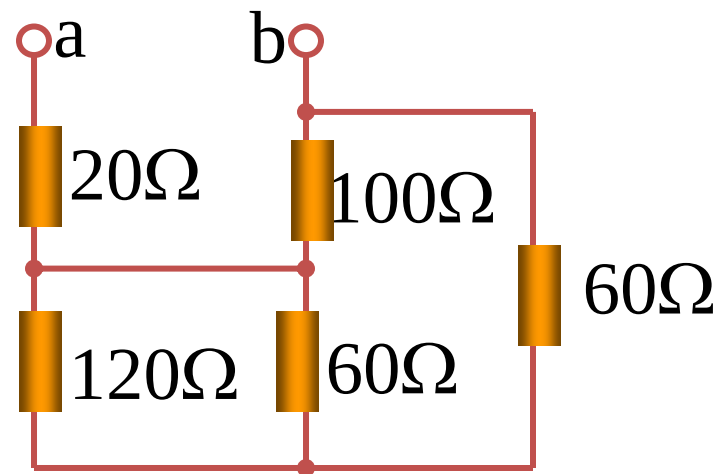
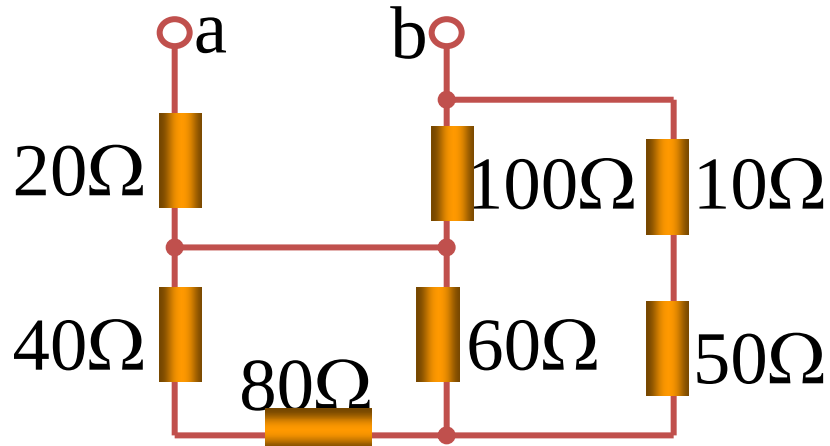
$$R_{cd} = (15 + 5) // 5 = 4\Omega$$

等效电阻针对端口而言

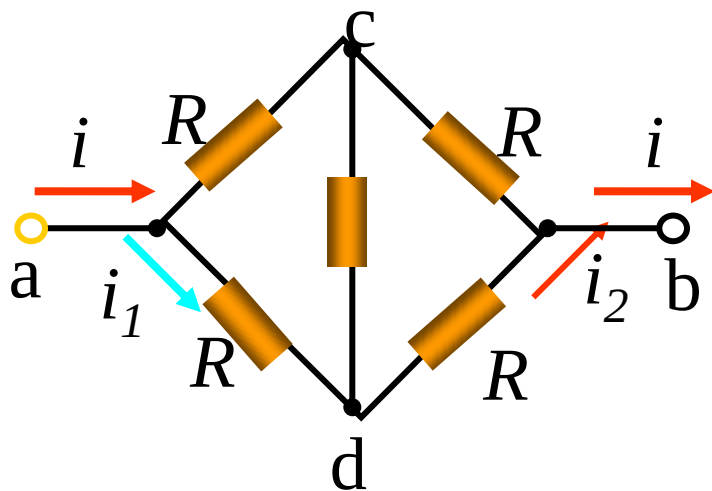
例2

求:  $R_{ab}$

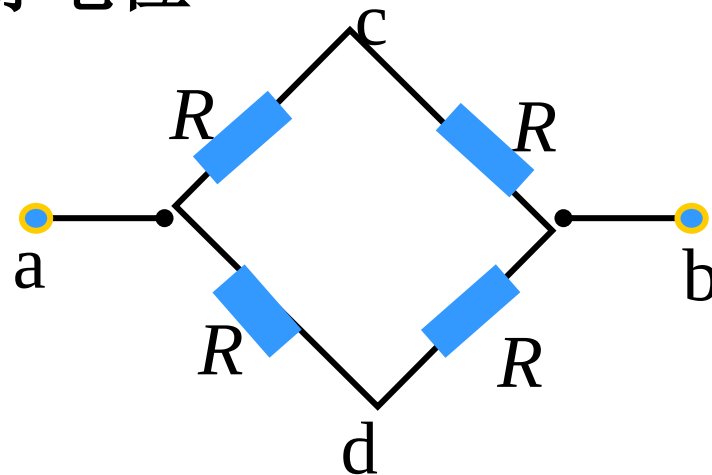
$$R_{ab} = 70\Omega$$



例3 求:  $R_{ab}$  对称电路 c、d等电位



短路

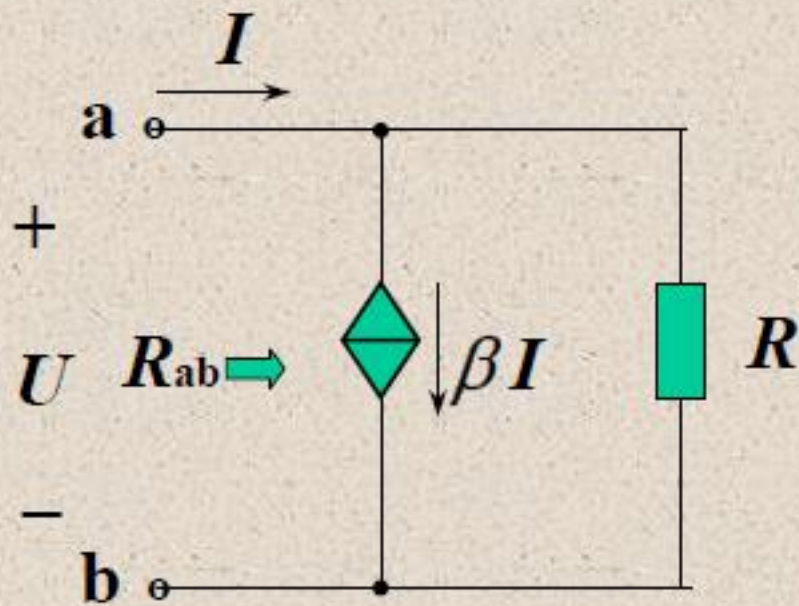


根据电流分配  $\Rightarrow i_1 = \frac{1}{2}i = i_2 \quad R_{ab} = R$

$$u_{ab} = i_1 R + i_2 R = \left(\frac{1}{2}i + \frac{1}{2}i\right)R = iR$$

$$R_{ab} = \frac{u_{ab}}{i} = R$$

例 3. 求 a,b 两端的入端电阻  $R_{ab}$  ( $\beta \neq 1$ )



加流求压法求  $R_{ab}$

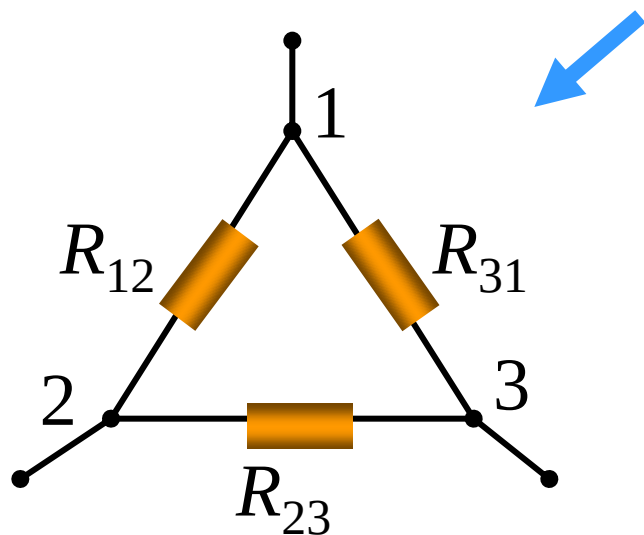
$$R_{ab} = U/I = (1 - \beta)R$$

## 2.4 电阻的Y形连接和 $\Delta$ 形连接的等效变换

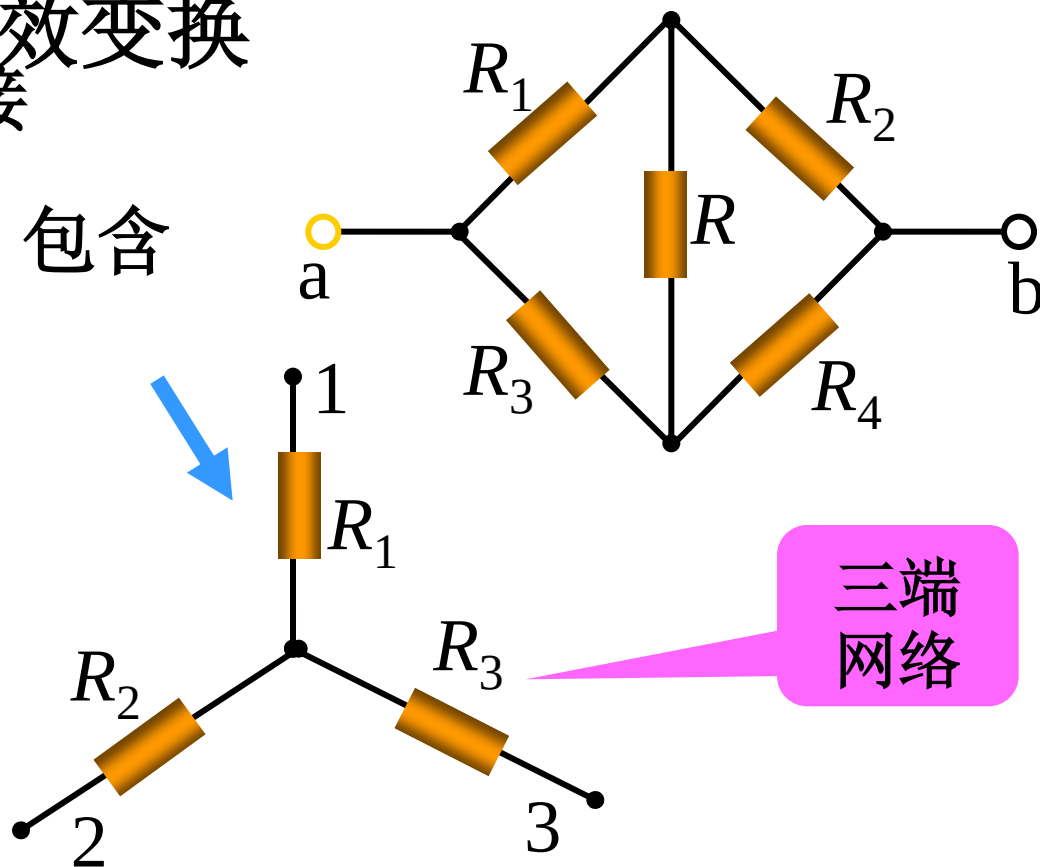
### 1. 电阻的 $\Delta$ 、Y形连接

等效变换

包含



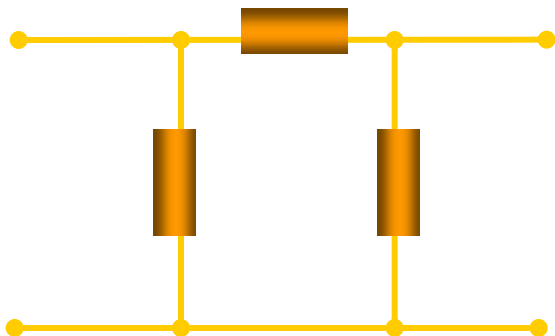
$\Delta$  形网络



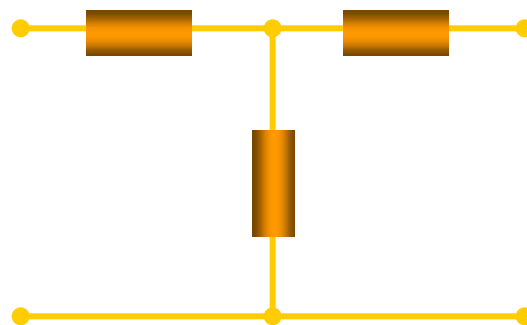
Y形网络

三端  
网络

$\Delta$  , Y 网络的变形:



$\pi$  型电路 ( $\Delta$  型)



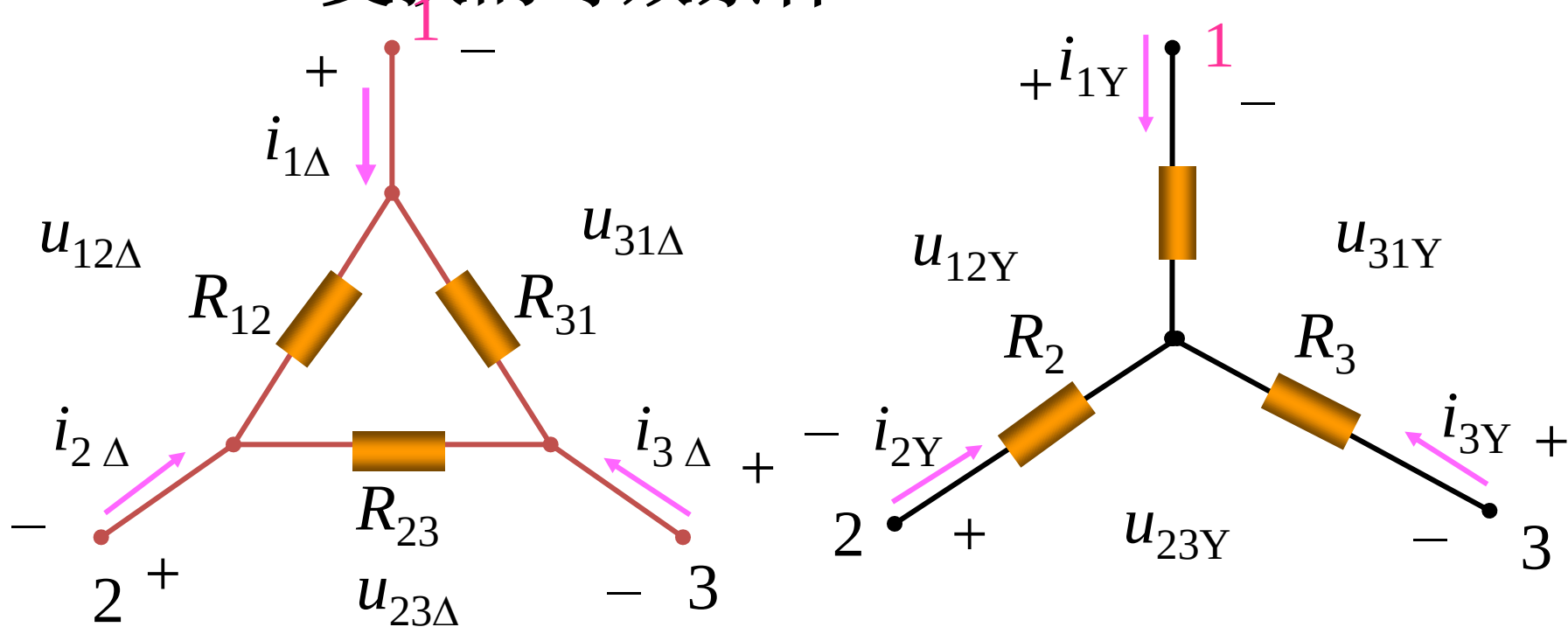
T 型电路 (Y、星型)



**注意**

这两个电路当它们的电阻满足一定的关系时，能够相互等效。

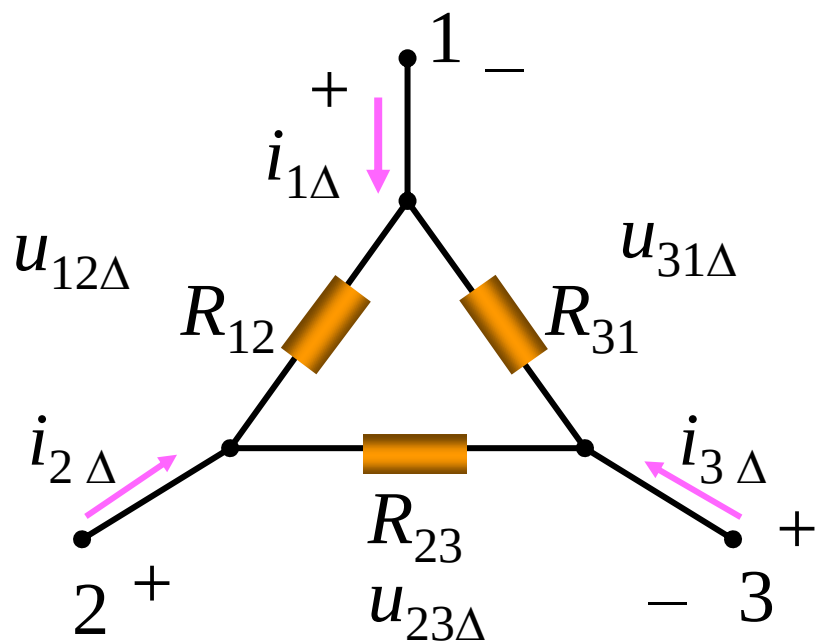
## 2. $\Delta$ —Y 变换的等效条件



等效条件:

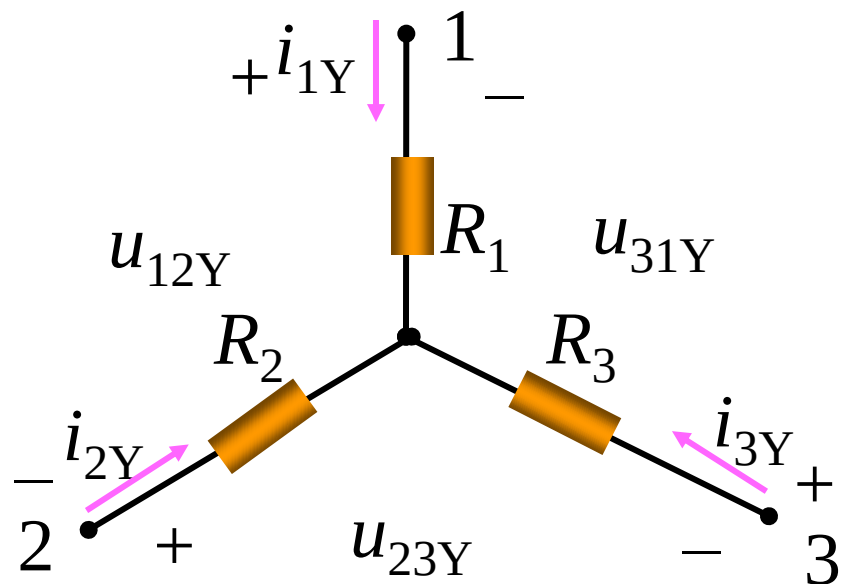
$$i_{1\Delta} = i_{1Y}, \quad i_{2\Delta} = i_{2Y}, \quad i_{3\Delta} = i_{3Y},$$

$$u_{12\Delta} = u_{12Y}, \quad u_{23\Delta} = u_{23Y}, \quad u_{31\Delta} = u_{31Y}$$



Δ接：用电压表示电流

$$\left. \begin{aligned} i_{1\Delta} &= u_{12\Delta} / R_{12} - u_{31\Delta} / R_{31} \\ i_{2\Delta} &= u_{23\Delta} / R_{23} - u_{12\Delta} / R_{12} \\ i_{3\Delta} &= u_{31\Delta} / R_{31} - u_{23\Delta} / R_{23} \end{aligned} \right\} (1)$$



Y接：用电流表示电压

$$\left. \begin{aligned} u_{12Y} &= R_1 i_{1Y} - R_2 i_{2Y} \\ u_{23Y} &= R_2 i_{2Y} - R_3 i_{3Y} \\ u_{31Y} &= R_3 i_{3Y} - R_1 i_{1Y} \\ i_{1Y} + i_{2Y} + i_{3Y} &= 0 \end{aligned} \right\} (2)$$

由式(2)解得：

$$i_{1Y} = \frac{u_{12Y}R_3 - u_{31Y}R_2}{R_1R_2 + R_2R_3 + R_3R_1}$$

$$i_{2Y} = \frac{u_{23Y}R_1 - u_{12Y}R_3}{R_1R_2 + R_2R_3 + R_3R_1}$$

$$i_{3Y} = \frac{u_{31Y}R_2 - u_{23Y}R_1}{R_1R_2 + R_2R_3 + R_3R_1}$$

$$(3) \quad \begin{aligned} i_{1\Delta} &= u_{12\Delta}/R_{12} - u_{31\Delta}/R_{31} \\ i_{2\Delta} &= u_{23\Delta}/R_{23} - u_{12\Delta}/R_{12} \end{aligned}$$

$$i_{3\Delta} = u_{31\Delta}/R_{31} - u_{23\Delta}/R_{23}$$

根据等效条件，比较式(3)与式(1)，得  
Y→Δ的变换条件：

$$\mathbf{R}_{12} = \mathbf{R}_1 + \mathbf{R}_2 + \frac{\mathbf{R}_1 \mathbf{R}_2}{\mathbf{R}_3}$$

$$\mathbf{R}_{23} = \mathbf{R}_2 + \mathbf{R}_3 + \frac{\mathbf{R}_2 \mathbf{R}_3}{\mathbf{R}_1}$$

$$\mathbf{R}_{31} = \mathbf{R}_3 + \mathbf{R}_1 + \frac{\mathbf{R}_3 \mathbf{R}_1}{\mathbf{R}_2}$$

$$G_{12} = \frac{G_1 G_2}{G_1 + G_2 + G_3}$$

$$\text{或 } G_{23} = \frac{G_2 G_3}{G_1 + G_2 + G_3}$$

$$G_{31} = \frac{G_3 G_1}{G_1 + G_2 + G_3}$$

类似可得到由 $\Delta \rightarrow Y$ 的变换条件：

$$\mathbf{G}_1 = \mathbf{G}_{12} + \mathbf{G}_{31} + \frac{\mathbf{G}_{12} \mathbf{G}_{31}}{\mathbf{G}_{23}}$$

$$\mathbf{G}_2 = \mathbf{G}_{23} + \mathbf{G}_{12} + \frac{\mathbf{G}_{23} \mathbf{G}_{12}}{\mathbf{G}_{31}}$$

$$\mathbf{G}_3 = \mathbf{G}_{31} + \mathbf{G}_{23} + \frac{\mathbf{G}_{31} \mathbf{G}_{23}}{\mathbf{G}_{12}}$$

或

$$\mathbf{R}_1 = \frac{\mathbf{R}_{12} \mathbf{R}_{31}}{\mathbf{R}_{12} + \mathbf{R}_{23} + \mathbf{R}_{31}}$$

$$\mathbf{R}_2 = \frac{\mathbf{R}_{23} \mathbf{R}_{12}}{\mathbf{R}_{12} + \mathbf{R}_{23} + \mathbf{R}_{31}}$$

$$\mathbf{R}_3 = \frac{\mathbf{R}_{31} \mathbf{R}_{23}}{\mathbf{R}_{12} + \mathbf{R}_{23} + \mathbf{R}_{31}}$$

简记方法:

$$R_Y = \frac{\Delta \text{相邻电阻乘积}}{\sum R_{\Delta}}$$

$\Delta$ 变Y

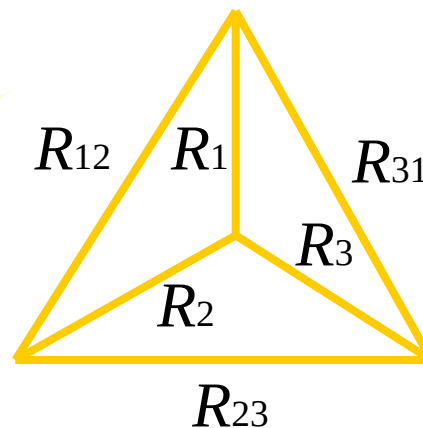
$$G_{\Delta} = \frac{Y \text{相邻电导乘积}}{\sum G_Y}$$

Y变 $\Delta$

特例：若三个电阻相等(对称)，则有

$$R_{\Delta} = 3R_Y$$

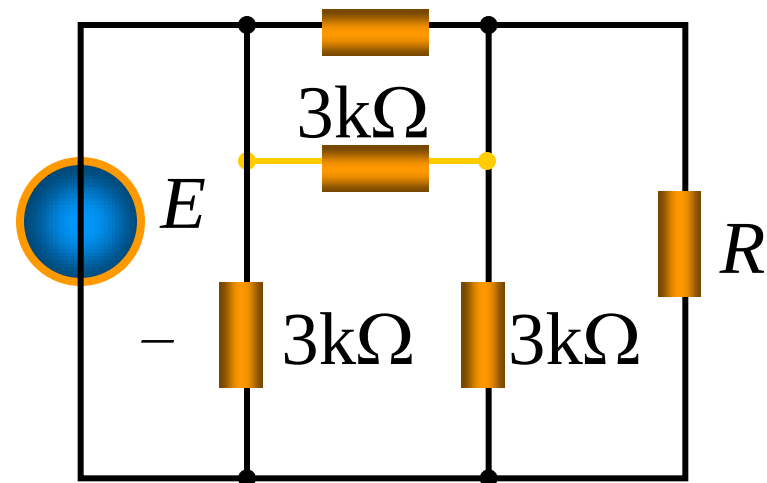
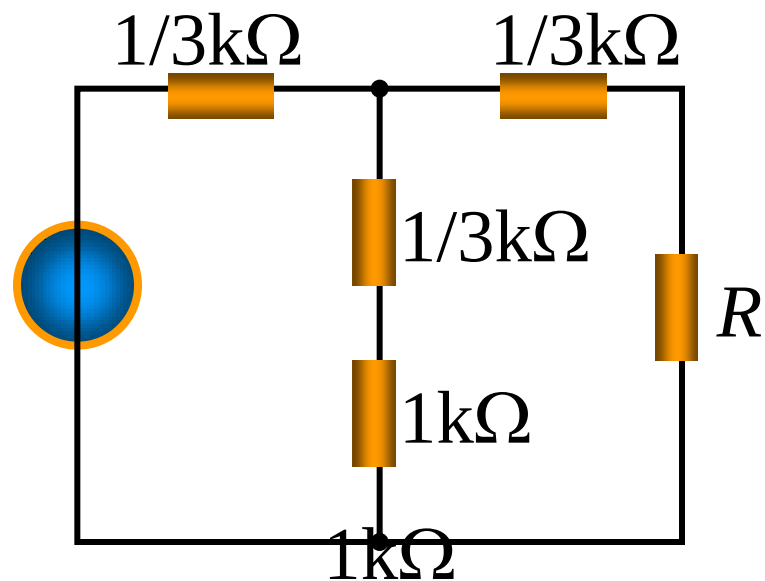
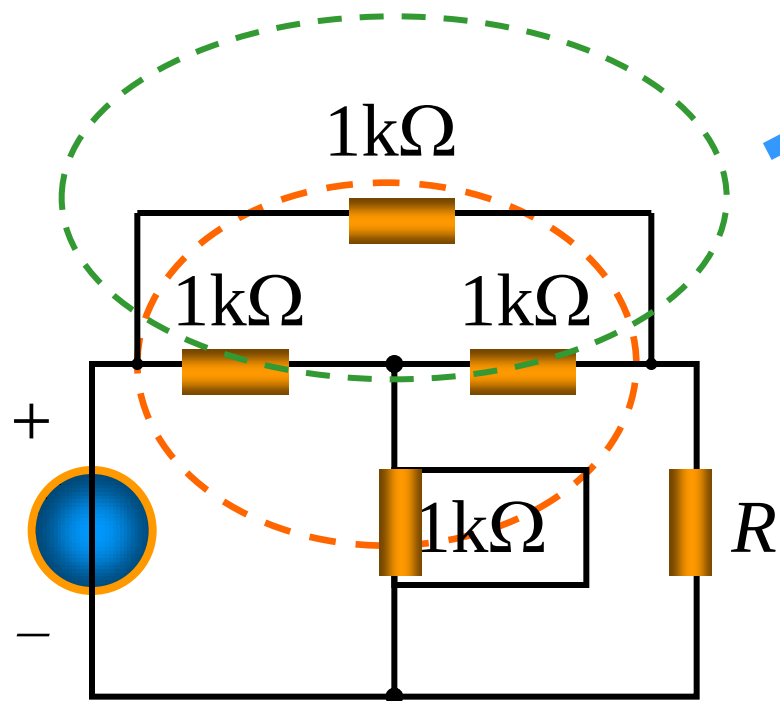
外大内小



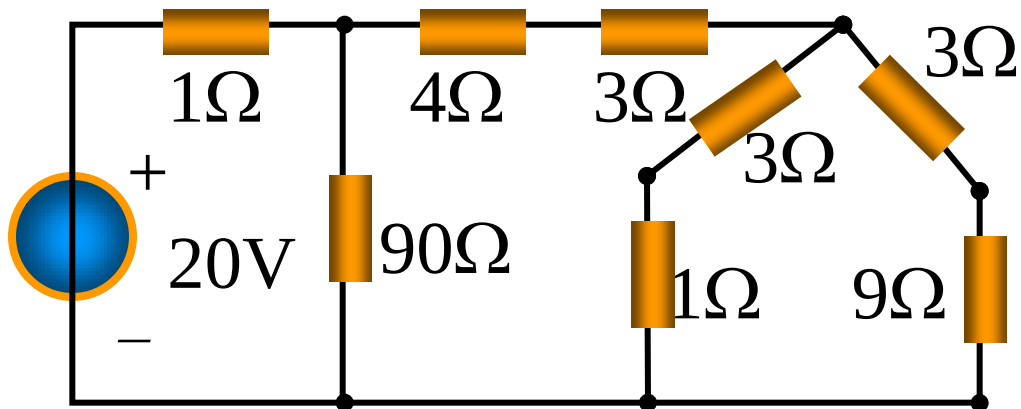
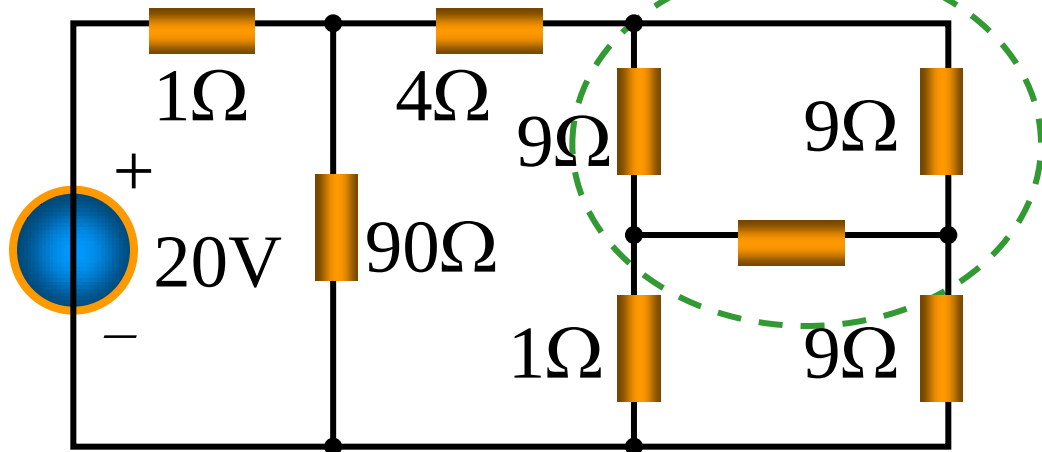


- ①等效对外部(端钮以外)有效, 对内不成立。
- ②等效电路与外部电路无关。
- ③用于简化电路

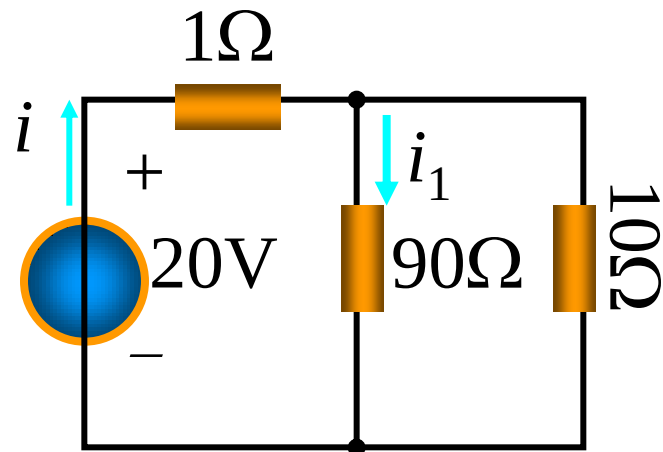
# 例1 桥 T 电路



例2 计算90Ω电阻吸收的功率



$$P = 90i_1^2 = 90 \times (0.2)^2 = 3.6 \text{ W}$$



$$R_{eq} = 1 + \frac{10 \times 90}{10 + 90} = 10 \Omega$$

$$i = 20 / 10 = 2 \text{ A}$$

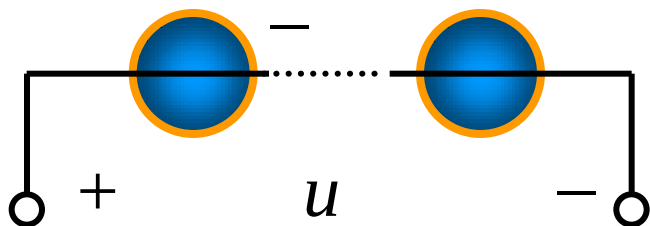
$$i_1 = \frac{10 \times 2}{10 + 90} = 0.2 \text{ A}$$

## 2.5 电压源、电流源的串联和并联

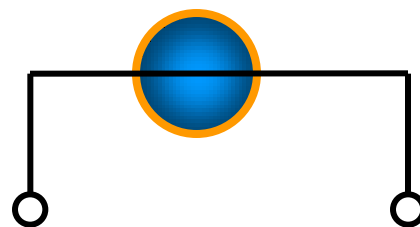
### 1. 理想电压源的串联和并联

注意参考方向

① 串联  $u = u_{s1} + u_{s2} = \sum u_{sk}$



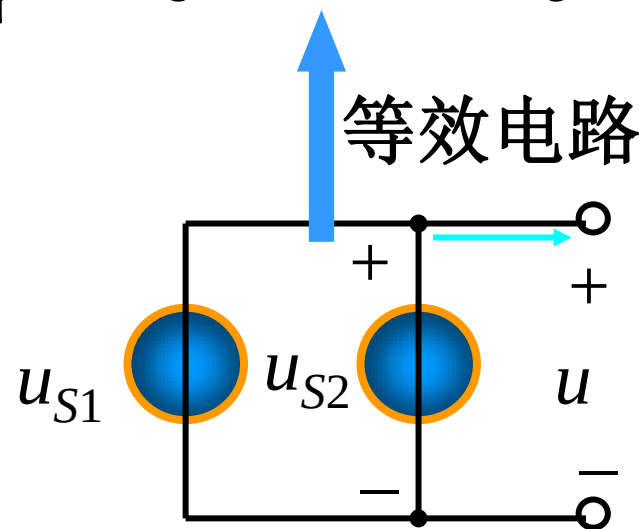
等效电路



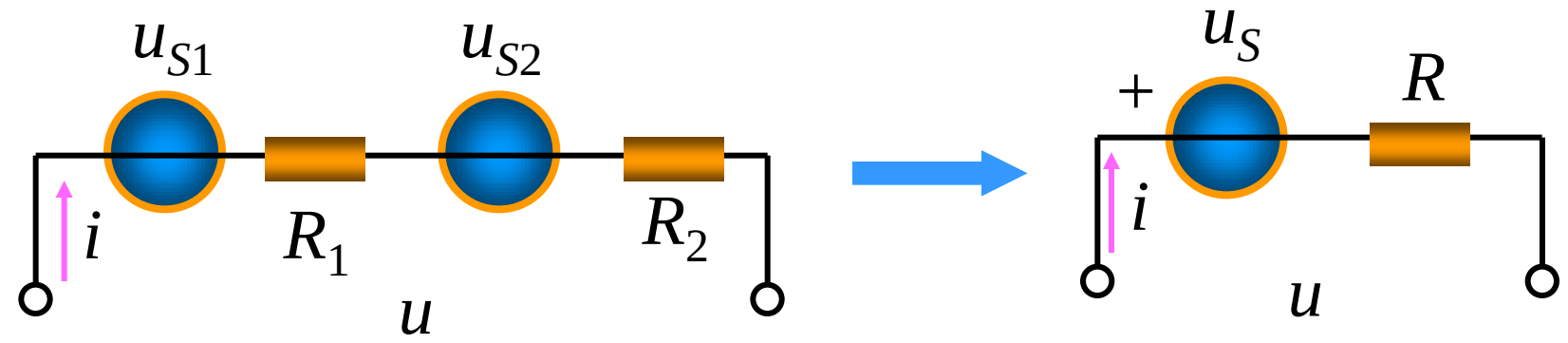
② 并联  $u = u_{s1} = u_{s2}$



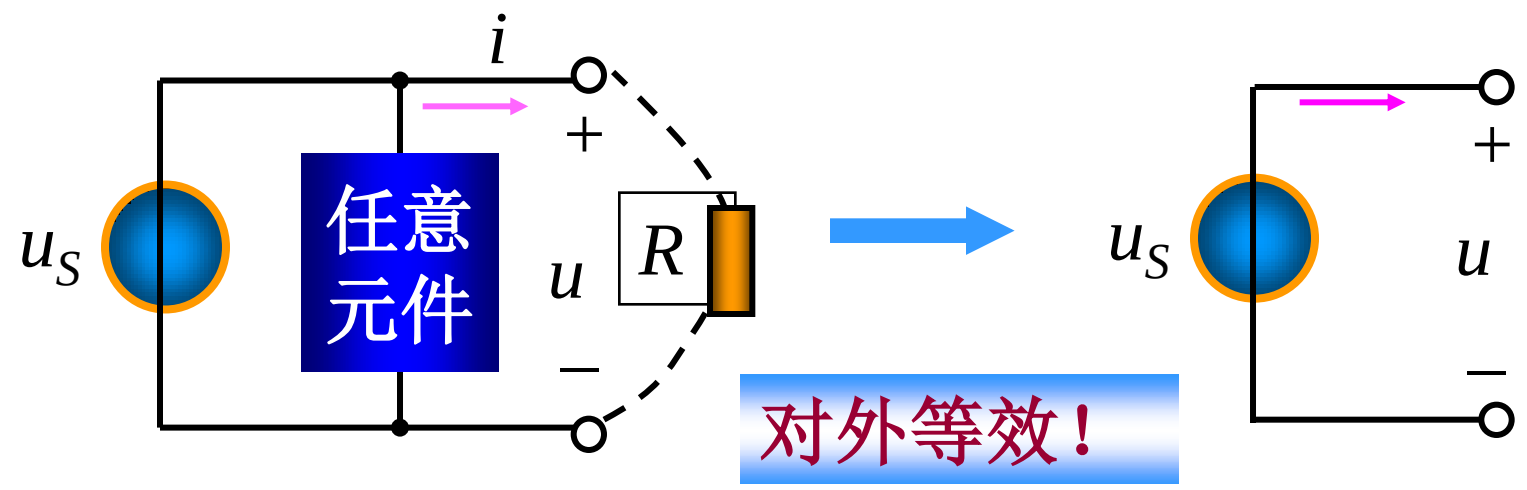
**注意** 相同电压源才能并联, 电源中的电流不确定。



③电压源与支路的串、并联等效



$$u = u_{s1} + R_1 i + u_{s2} + R_2 i = (u_{s1} + u_{s2}) + (R_1 + R_2) i = u_S + R i$$

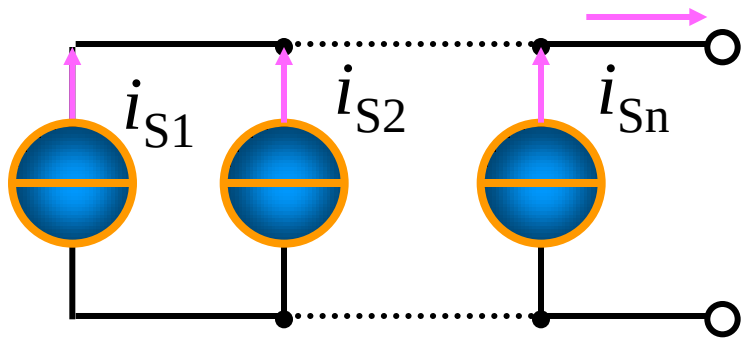


## 2. 理想电流源的串联并联

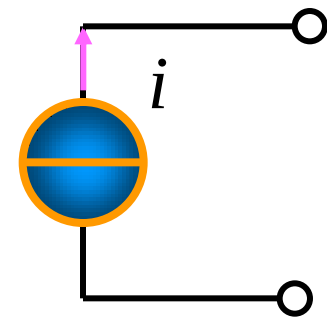
注意参考方向

① 并联

$$i = i_{s1} + i_{s2} + \cdots + i_{sn} = \sum i_{sk}$$



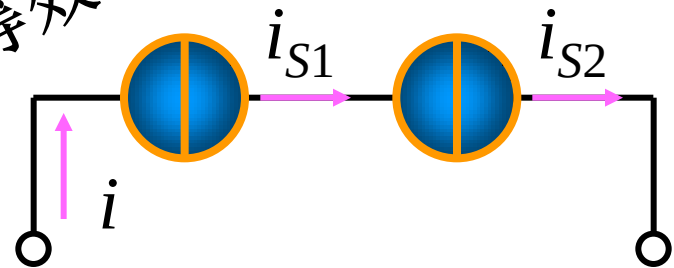
等效电路



② 串联

$$i = i_{s1} = i_{s2}$$

等效电路

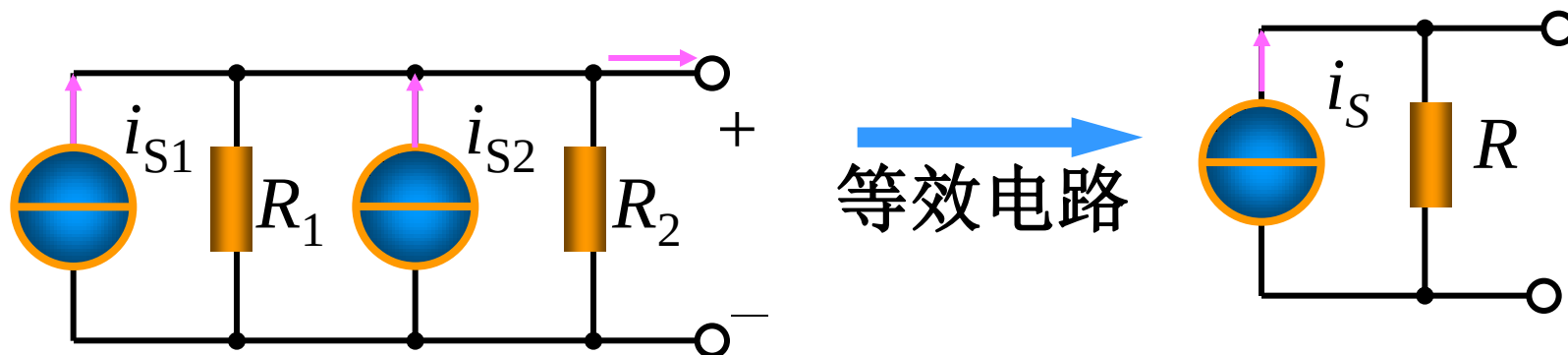


注意

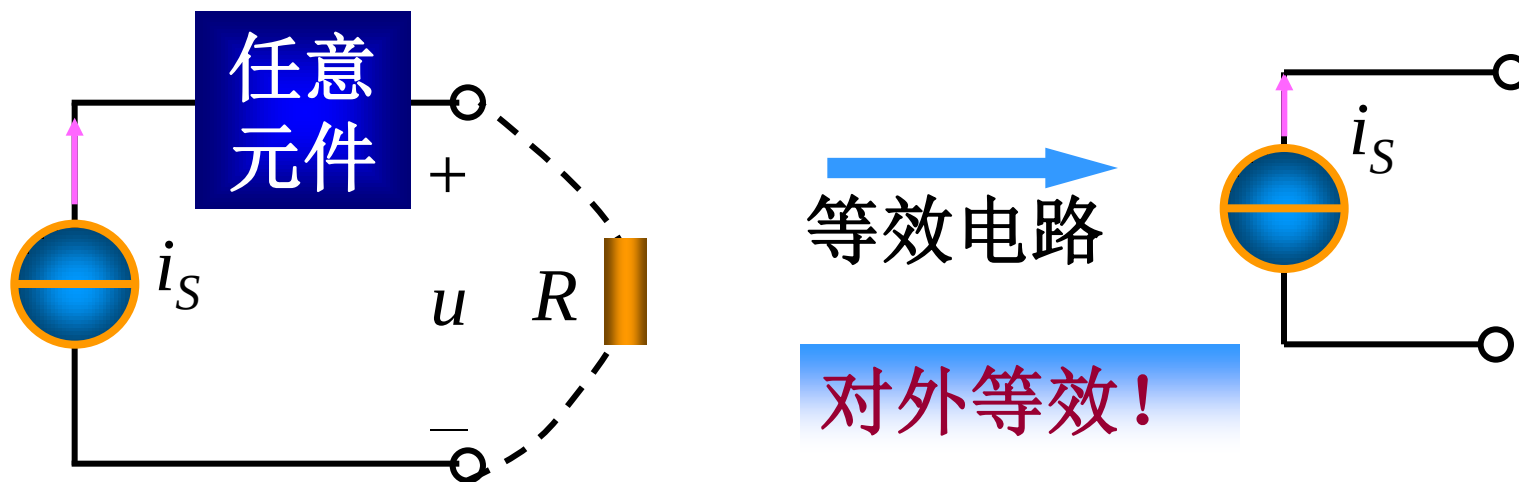
相同的理想电流源才能串联，每个电

流源的端电压不能确定。

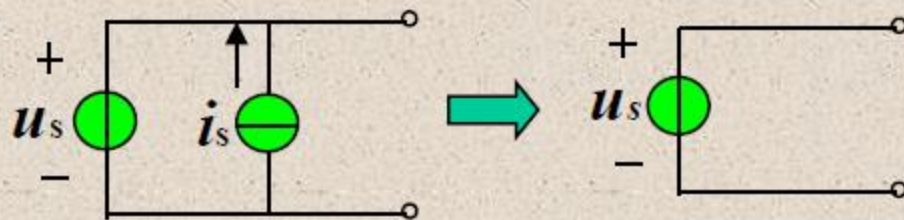
### 3. 电流源与支路的串、并联等效



$$i = i_{s1} - u/R_1 + i_{s2} - u/R_2 = i_{s1} + i_{s2} - (1/R_1 + 1/R_2)u = i_s - u/R$$



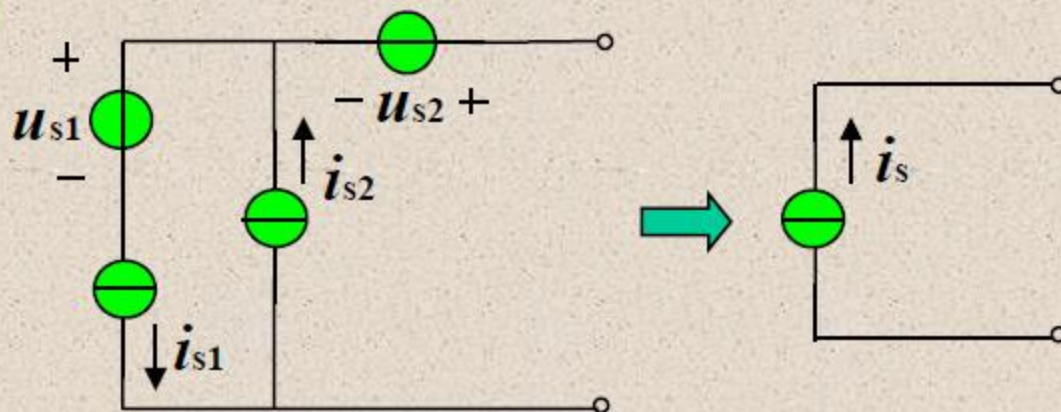
例1:



例2:



例3:

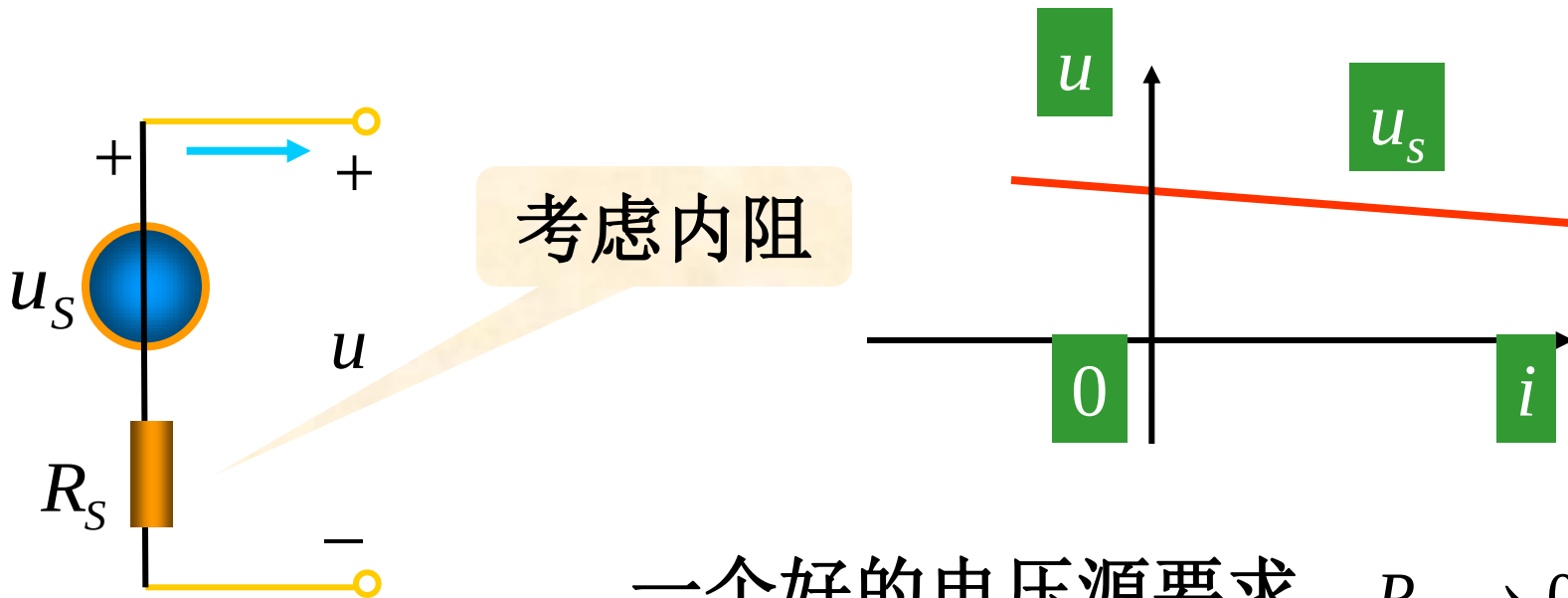


$$i_s = i_{s2} - i_{s1}$$



## 2.6 实际电源的两种模型及其等效变换

1. 实际电压源 伏安特性:  $u = u_S - R_S i$



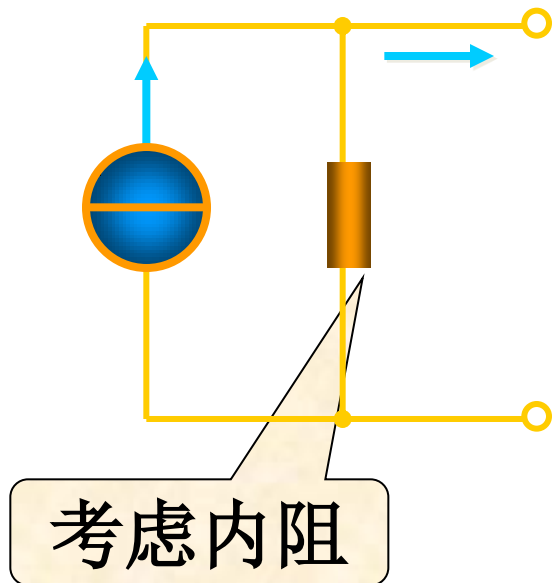
一个好的电压源要求  $R_S \rightarrow 0$



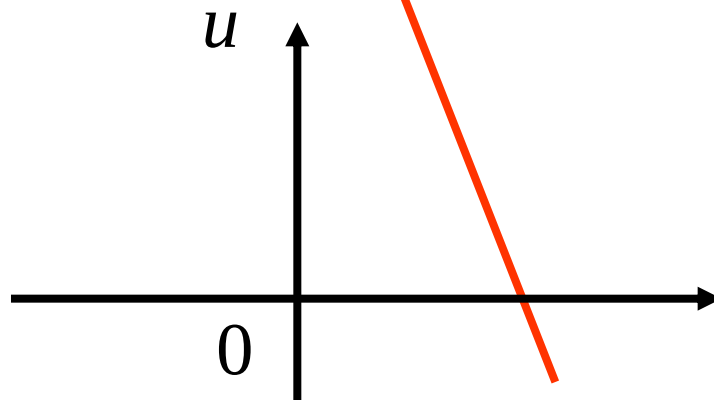
**注意**

实际电压源也不允许短路。因其内阻小，若短路，电流很大，可能烧毁电源。

## 2. 实际电流源



伏安特性:  $i = i_s - \frac{u}{R_s}$



一个好的电流源要求

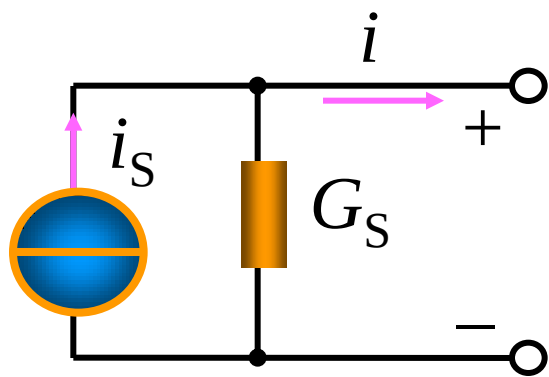


**注意**

实际电流源也不允许开路。因其内阻大，若开路，电压很高，可能烧毁电源。

### 3. 电压源和电流源的等效变换

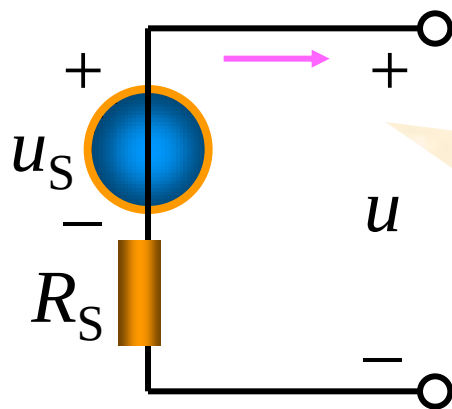
实际电压源、实际电流源两种模型可以进行等效变换，所谓的等效是指端口的电压、电流在转换过程中保持不变。



实际  
电流  
源

端口特性  $i = i_S - G_S u$

$$i_S = u_S / R_S$$
$$G_S = 1 / R_S$$



实际  
电压  
源

$$u = u_S - R_S i$$

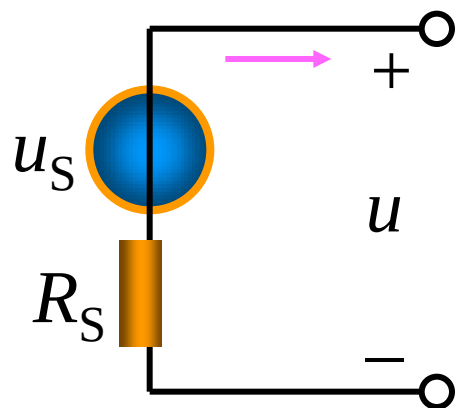
$$i = u_S / R_S - u / R_S$$

比较可得等效条件

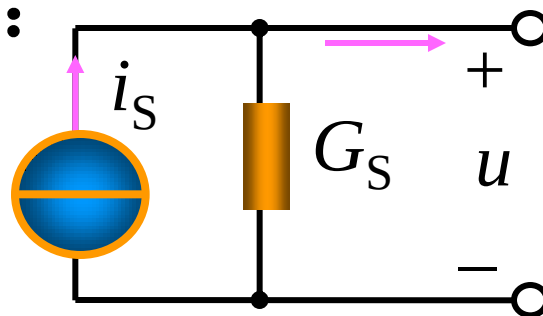


小结

电压源变换为电流源:

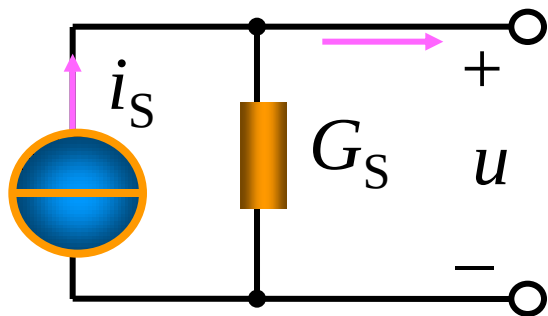


转换

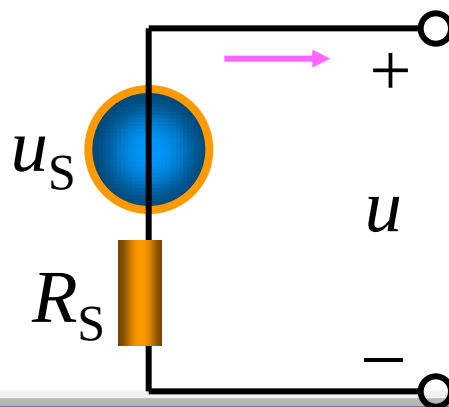


$$i_s = u_s / R_s, \quad G_s = 1 / R_s$$

电流源变换为电压源:



转换



$$u_s = i_s / G_s, \quad R_s = 1 / G_s$$



注意

## ①变换关系

数值关系

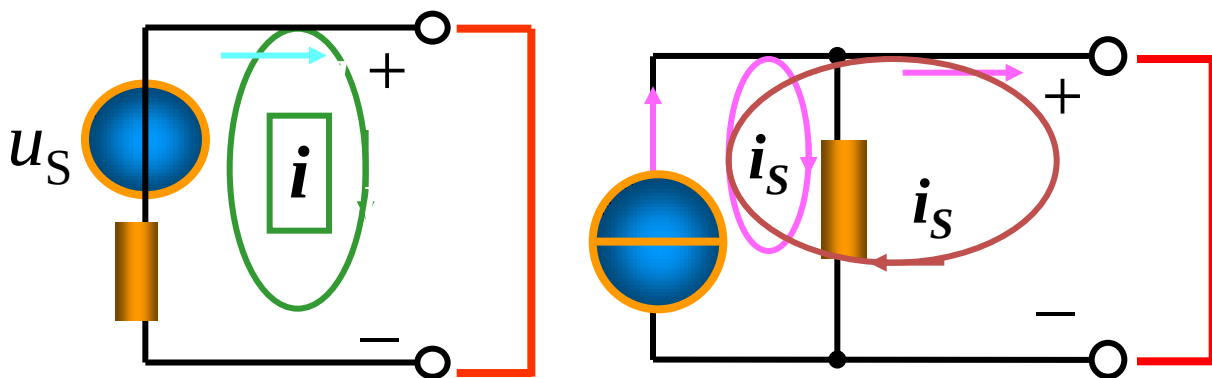
方向：电流源电流方向与电压源电压方向相反。

②等效是对外部电路等效，对内部电路是不等效的。

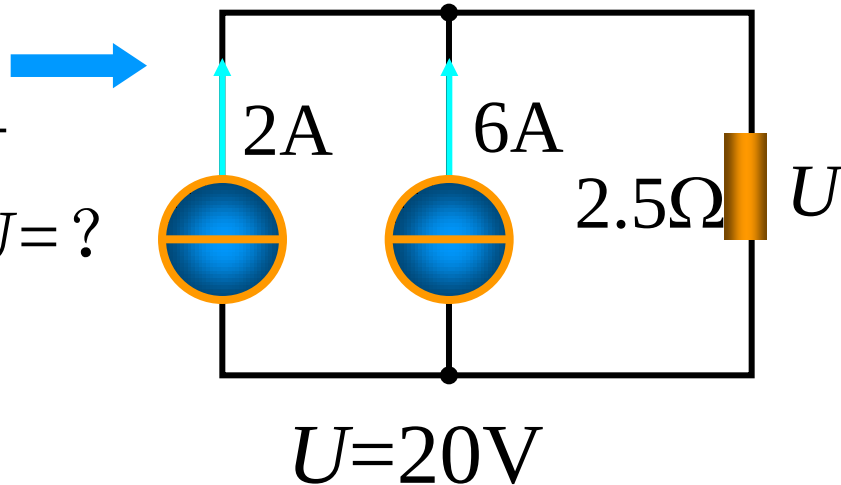
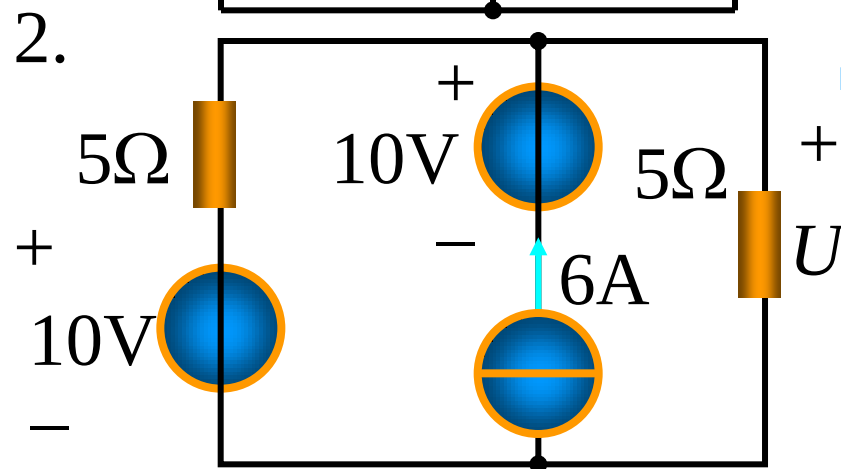
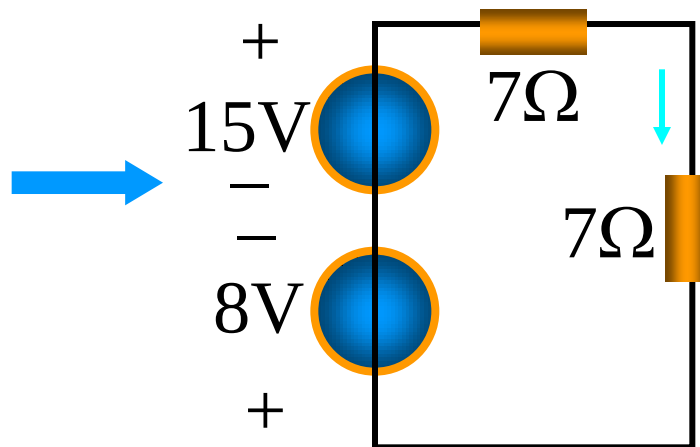
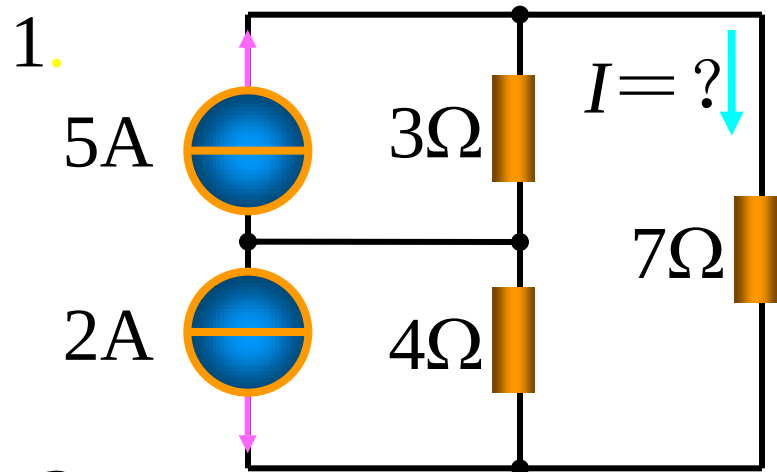
表现在

- 电压源开路， $R_S$ 上无电流流过
- 电压源短路， $R_S$ 上有电流；
- 电流源开路， $G_S$ 上有电流流过。
- 电流源短路， $G_S$ 上无电流。

③理想电压源与理想电流源不能相互转换。

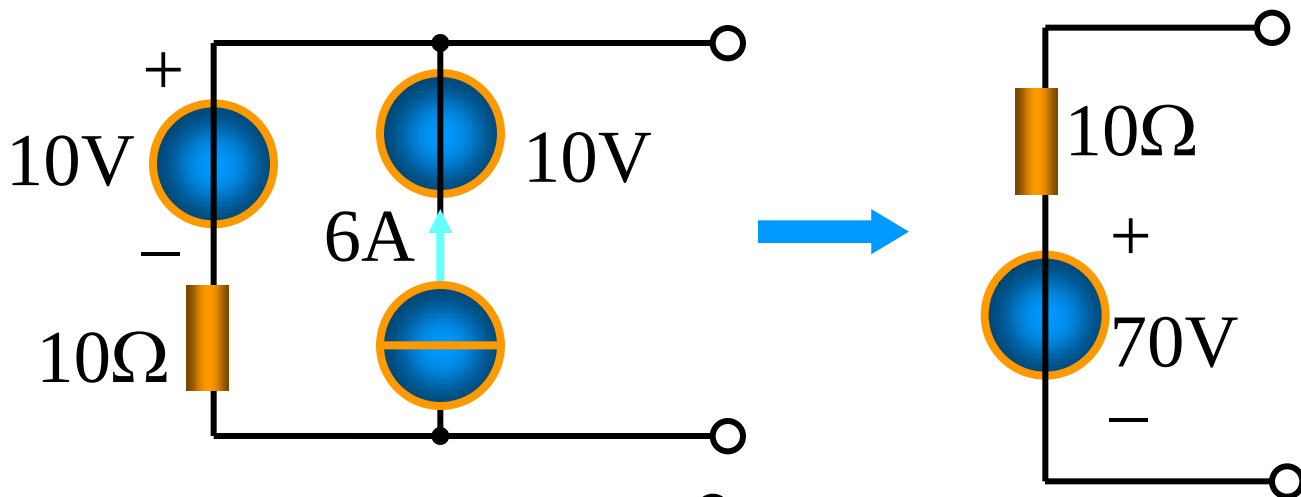


# 例1 利用电源转换简化电路计算

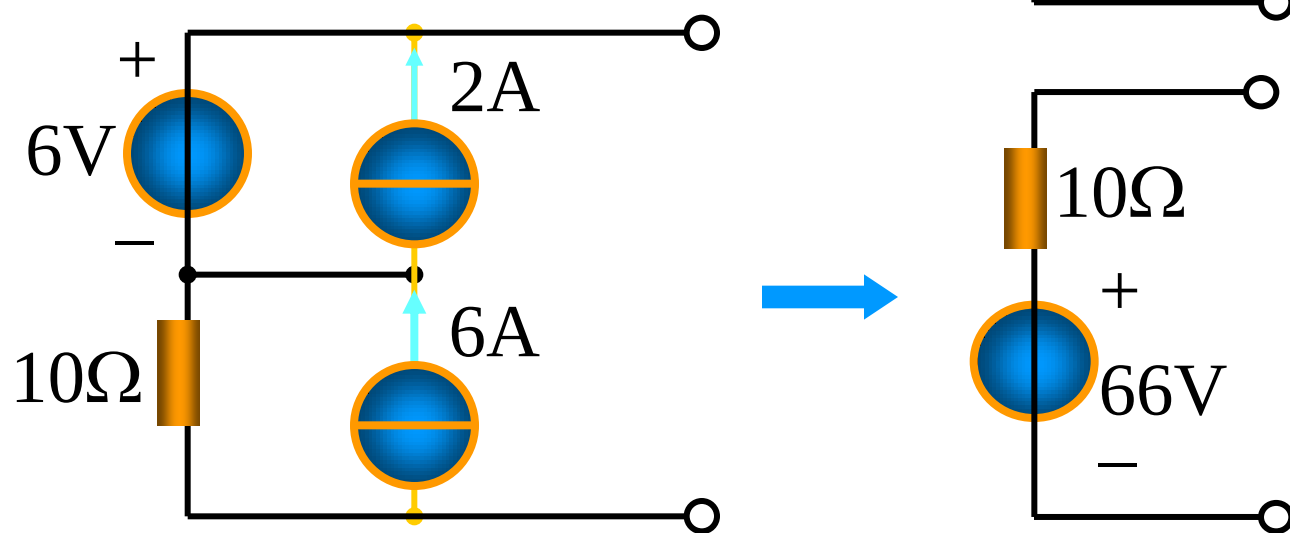


## 例2 把电路转换成一个电压源和一个电阻的串联

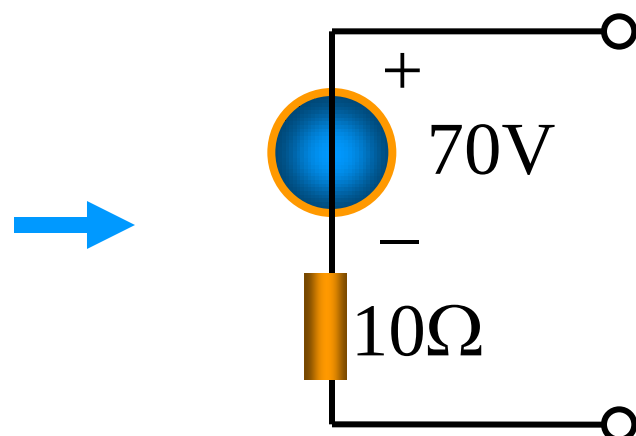
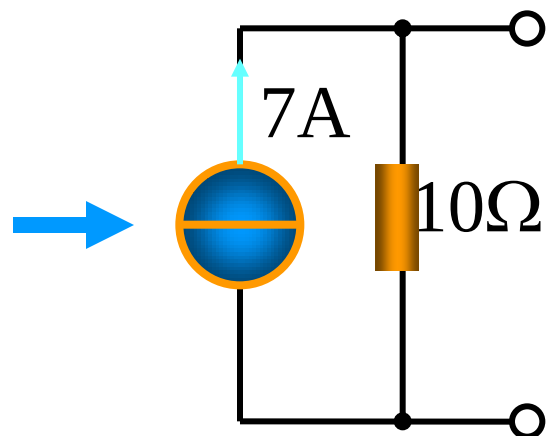
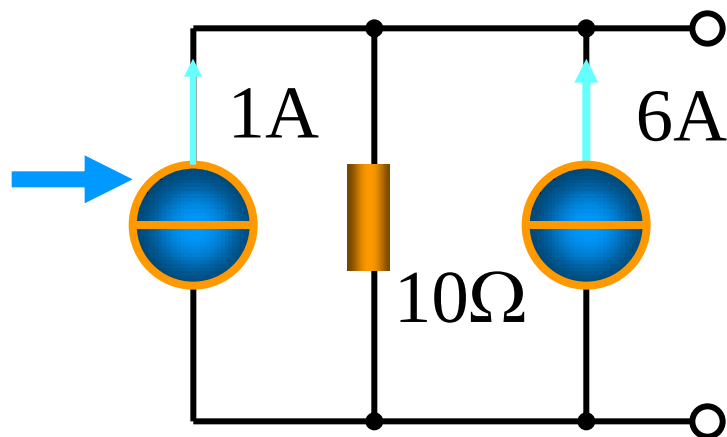
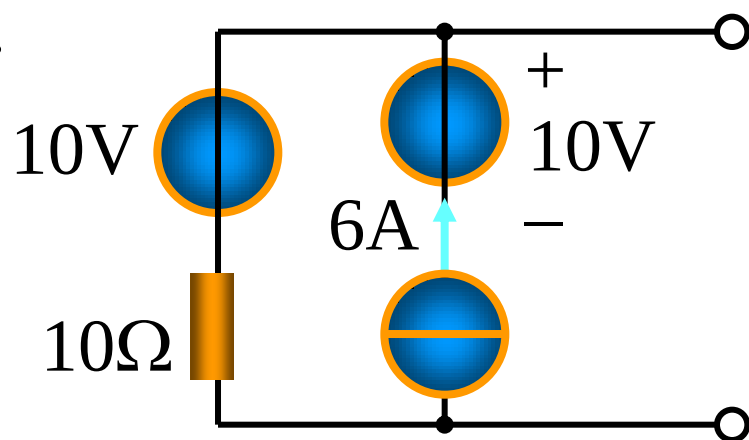
1.



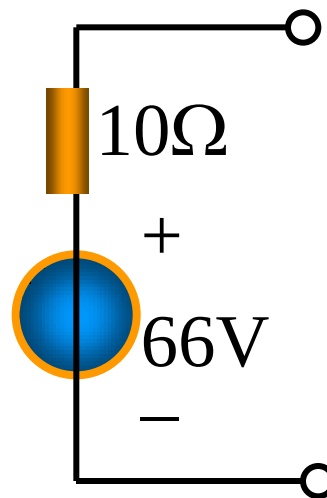
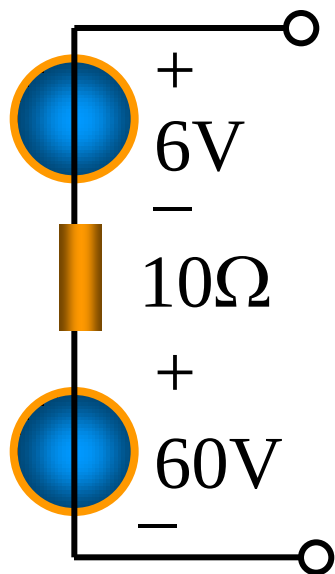
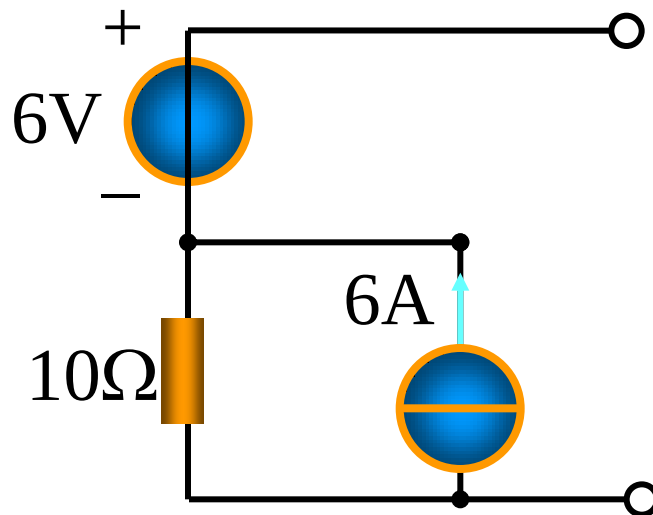
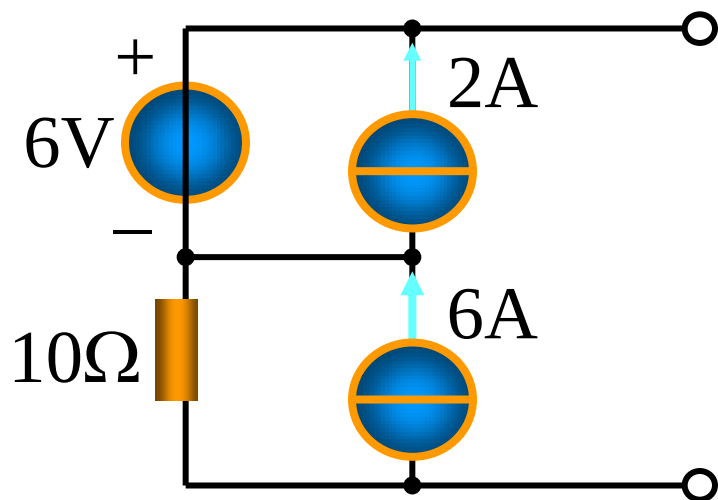
2.



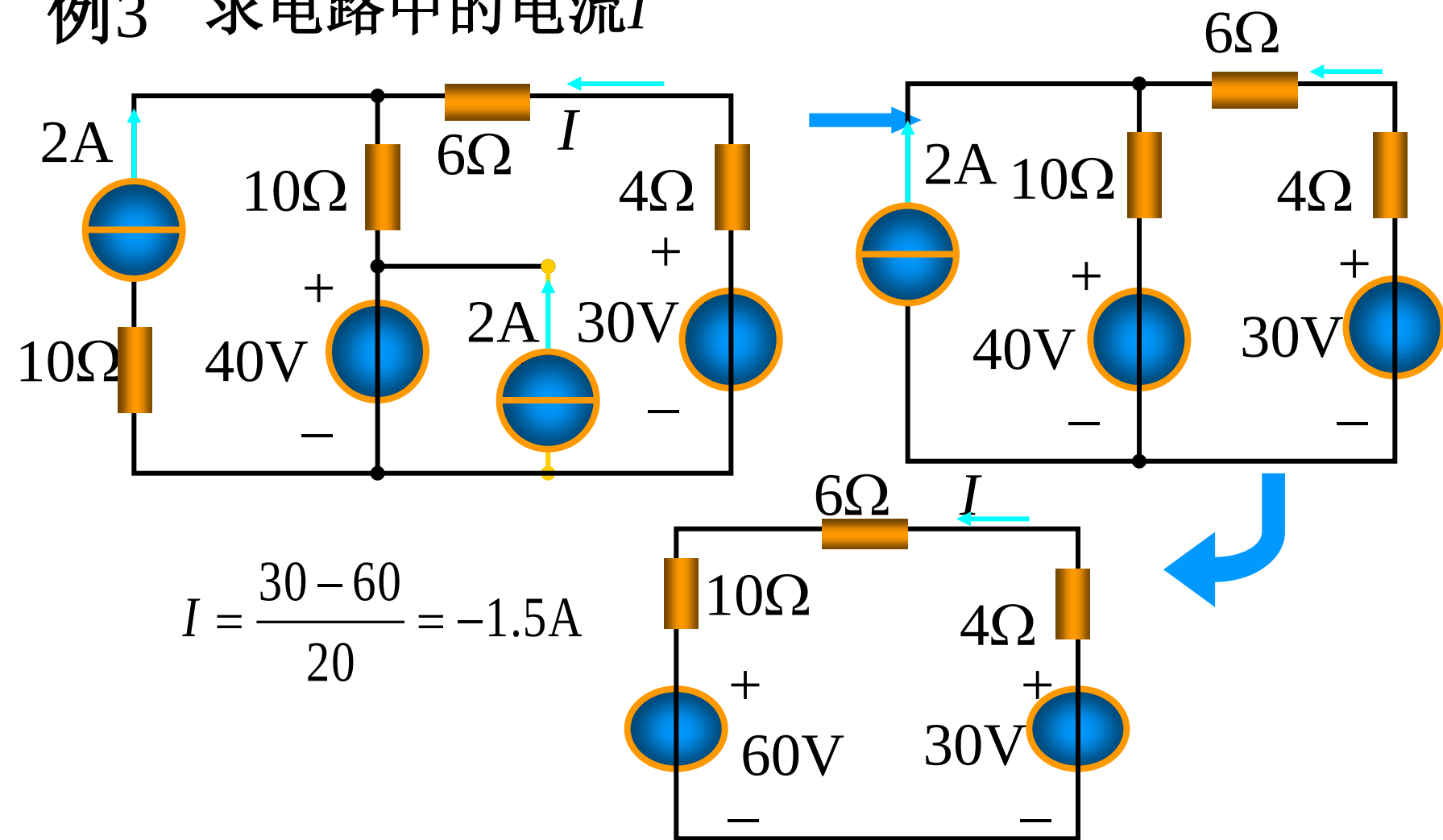
1.



2.



### 例3 求电路中的电流 $I$



例4 求电流  $i_1$

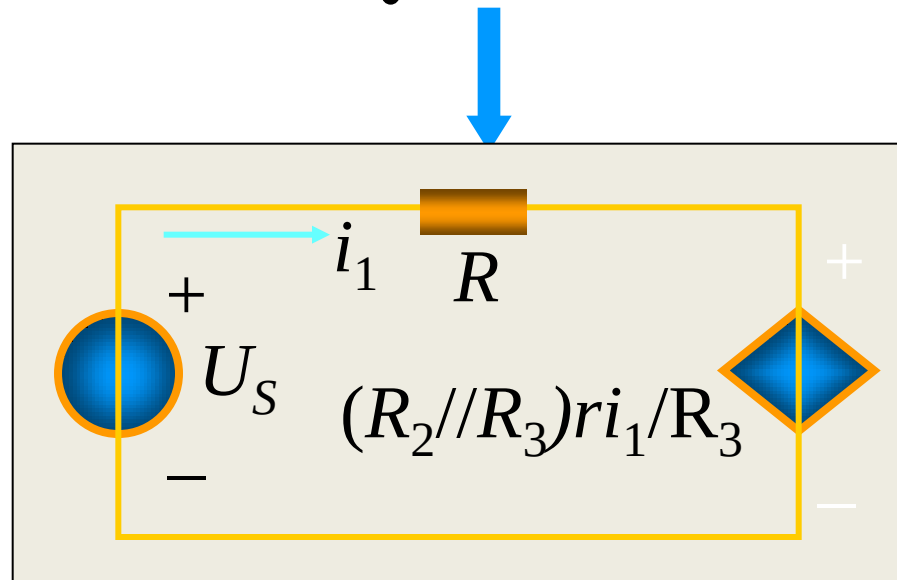
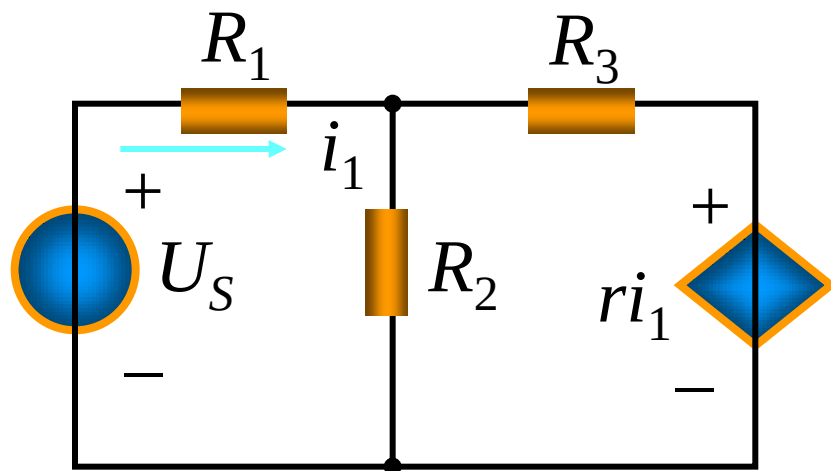
$$R = R_1 + \frac{R_2 R_3}{R_2 + R_3}$$

$$Ri_1 + (R_2 // R_3)ri_1 / R_3 = U_s$$

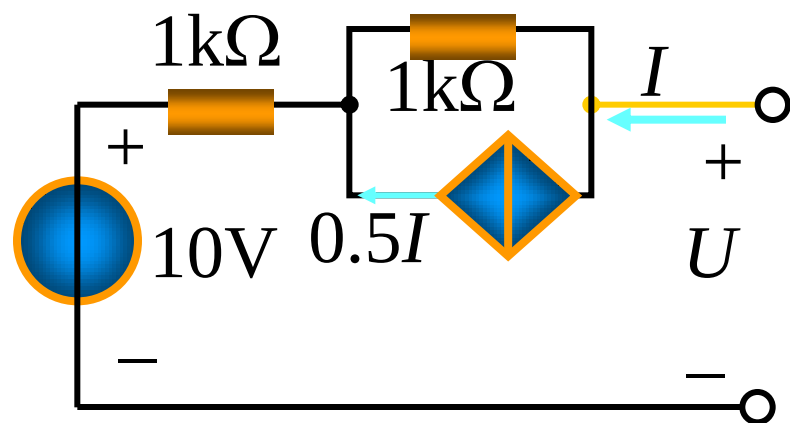
$$i_1 = \frac{U_s}{R + (R_2 // R_3)r / R_3}$$



**注意** 受控源和独立源一样可以进行电源转换；转换过程中注意不要丢失控制量。

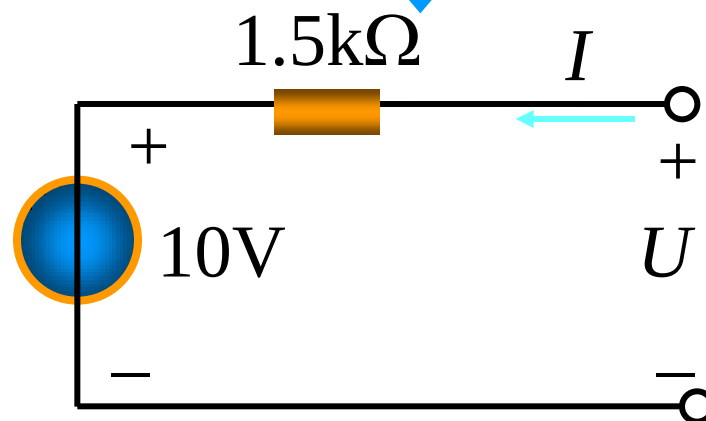
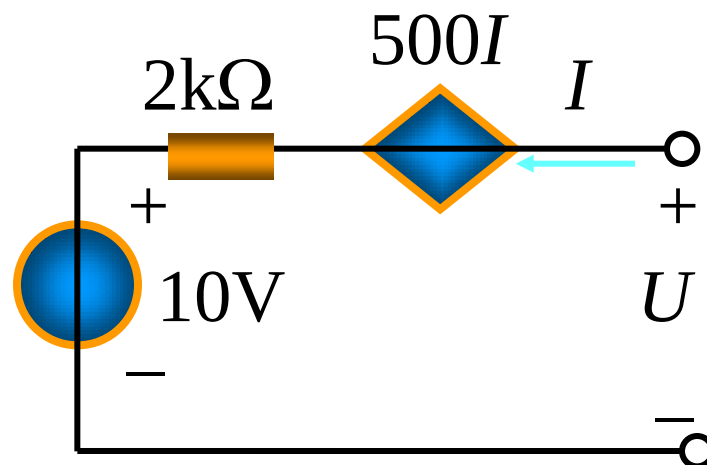


# 例5 把电路转换成一个电压源和一个电阻的串联



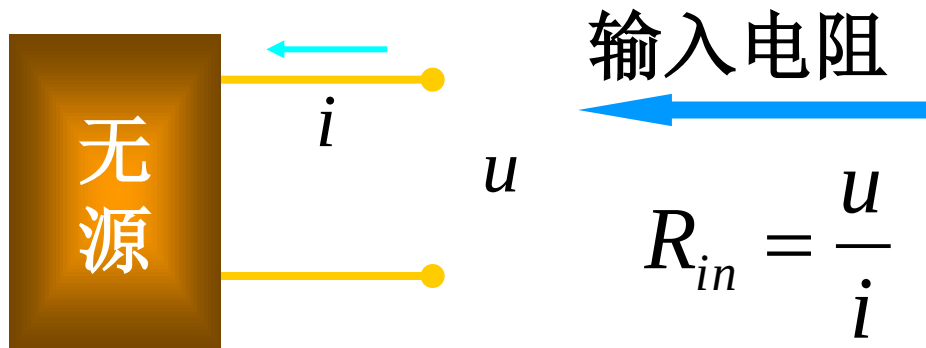
$$U = -500I + 2000I + 10$$

$$= 1500I + 10$$



## 2.7 输入电阻

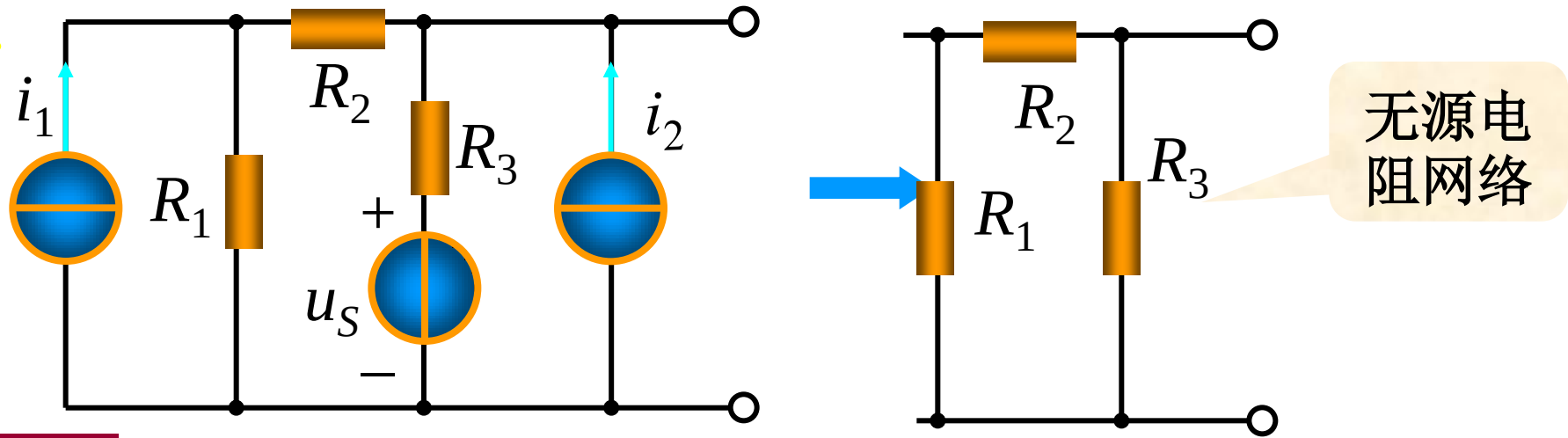
### 1. 定义



### 2. 计算方法

- ①如果一端口内部仅含电阻，则应用电阻的串、并联和 $\Delta$ —Y变换等方法求它的等效电阻；
- ②对含有受控源和电阻的两端电路，用电压、电流法求输入电阻，即在端口加电压源，求得电流，或在端口加电流源，求得电压，得其比值。

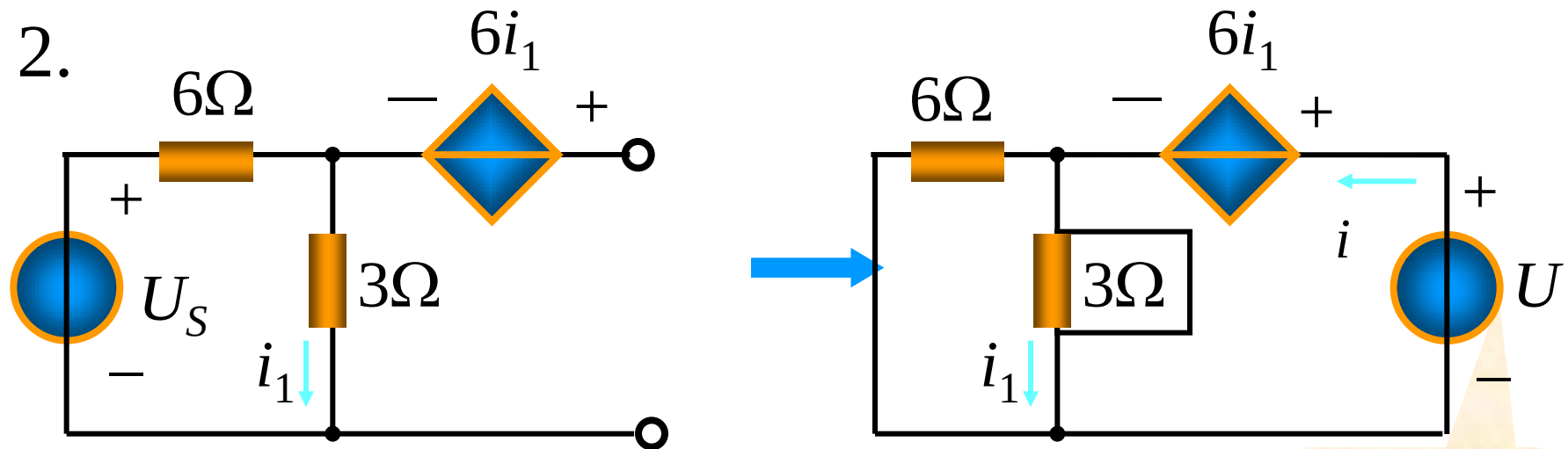
例 1. 计算下例一端口电路的输入电阻



**解**

先把有源网络的独立源置零：电压源短路；  
电流源开路，再求输入电阻。

$$R_{in} = (R_1 + R_2) // R_3$$



$$i = i_1 + \frac{3i_1}{6} = 1.5i_1$$

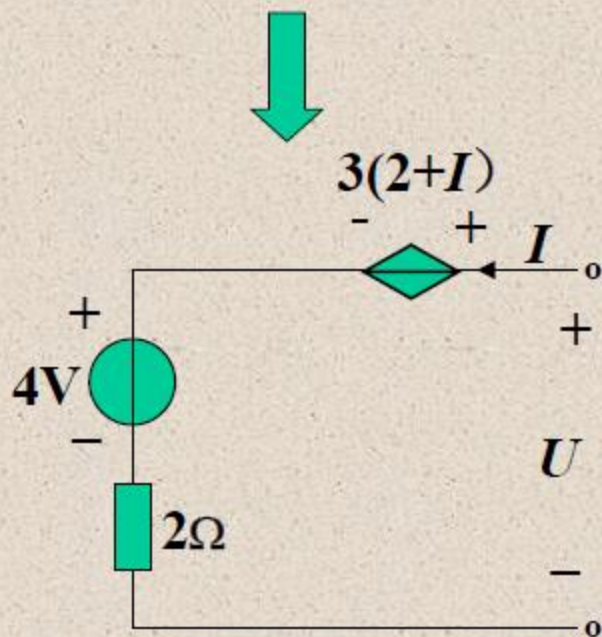
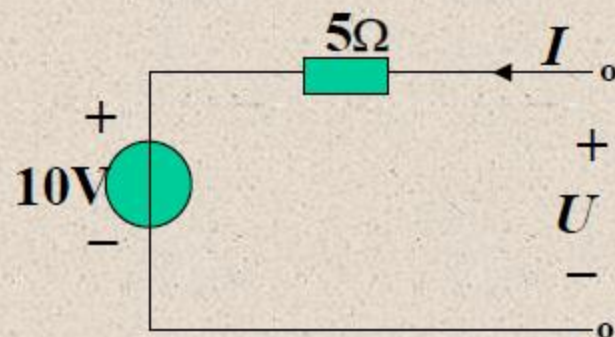
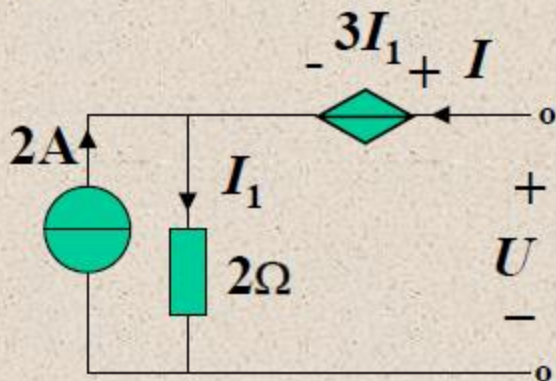
$$U = 6i_1 + 3i_1 = 9i_1$$

$$R_{in} = \frac{U}{i} = \frac{9i_1}{1.5i_1} = 6\Omega$$

外加电  
压源

例5.

$$U = 3I_1 + 2I_1 = 5I_1 = 5(2 + I) = 10 + 5I$$



$$U = 3(2 + I) + 4 + 2I = 10 + 5I$$